



Chulalongkorn University

Here for Free Lunch

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template from KACTL

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Template (1)

template.cpp31 lines

```
#pragma once
#include <bits/stdc++.h>
#define sz(x) (int)(x).size()
#define all(x) (x).begin(), (x).end()

using namespace std;

typedef long long ll;
typedef double db;
typedef long double ld;
typedef pair<int, int> pii;
typedef pair<ll, ll> pll;

template<typename T> bool ckmin(T &a, const T &b) { return b < a ? a = b, 1 : 0; }
template<typename T> bool ckmax(T &a, const T &b) { return a < b ? a = b, 1 : 0; }

mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());

const char nl = '\n';
const int INF = 0x3fffffff;
const int MOD=1000000007;
// const int MOD = 998244353;
const ll LINF = 0xffffffffffffffff;
const db DINF = numeric_limits<db>::infinity();
const db EPS = 1e-9;
const db PI = acos(db(-1));
```

1

```
signed main(){
    ios_base::sync_with_stdio(0); cin.tie(NULL);
    return 0;
}
```

1

c.sh2 lines

```
g++ -std=gnu++2a -Wall $1 -o a.out
./a.out
```

2

6

Mathematics (2)

2.1 Goldbatch’s Conjecture

- Even number can be written in sum of two primes (Up to 1e12)
- Range of N^{th} prime and $N + 1^{th}$ prime will be less than or equal to 300 (Up to 1e12)

2.2 Divisibility

Number of divisors of N is given by $\prod_{i=1}^k (a_i + 1)$ where $N = \prod_{i=1}^k p_i^{a_i}$ and p_i are prime factors of N .

Combinatorial (3)

3.1 Permutations

3.1.1 Factorial

n	1	2	3	4	5	6	7	8	9	10
$n!$	1	2	6	24	120	720	5040	40320	362880	3628800
n	11	12	13	14	15	16	17			
$n!$	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14			
n	20	25	30	40	50	100	150	171		
$n!$	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL_MAX		

IntPerm.h

Description: Permutation -> integer conversion. (Not order preserving). Integer -> permutation can use a lookup table.

Time: $\mathcal{O}(n)$

044568, 6 lines

```
int permToInt(vi &v){
    int use = 0, i = 0, r = 0;
    for (int x : v) r = r * ++i + __builtin_popcount(use & -(1 << x)),
        use |= 1 << x; // (note: minus, not ~!)
    return r;
}
```

3.1.2 Cycles

Let $g_S(n)$ be the number of n -permutations whose cycle lengths all belong to the set S . Then

$$\sum_{n=0}^{\infty} g_S(n) \frac{x^n}{n!} = \exp \left(\sum_{n \in S} \frac{x^n}{n} \right)$$

3.1.3 Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n) = (n-1)(D(n-1) + D(n-2)) = nD(n-1) + (-1)^n = \left\lfloor \frac{n!}{e} \right\rfloor$$

3.1.4 Burnside’s lemma

Given a group G of symmetries and a set X , the number of elements of X up to symmetry equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).

If $f(n)$ counts “configurations” (of some sort) of length n , we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(\gcd(n, k)) = \frac{1}{n} \sum_{k|n} f(k) \phi(n/k).$$

3.2 Partitions and subsets

3.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1, \quad p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k - 1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	$\sim 2\text{e}5$	$\sim 2\text{e}8$

3.2.2 Lucas’ Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + \dots + n_1 p + n_0$ and $m = m_k p^k + \dots + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^k \binom{n_i}{m_i} \pmod{p}$.

3.2.3 Binomials

multinomial.h

Description: Computes $\binom{k_1 + \dots + k_n}{k_1, k_2, \dots, k_n} = \frac{(\sum k_i)!}{k_1! k_2! \dots k_n!}$.

a0a312, 6 lines

```
ll multinomial(vi& v) {
    ll c = 1, m = v.empty() ? 1 : v[0];
    rep(i, 1, sz(v)) rep(j, 0, v[i])
        c = c * ++m / (j+1);
    return c;
}
```

3.3 General purpose numbers

3.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t - 1}$ (FFT-able). $B[0, \dots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \dots]$

Sums of powers:

$$\sum_{i=1}^n n^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k \cdot (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\sum_{i=m}^\infty f(i) = \int_m^\infty f(x)dx - \sum_{k=1}^\infty \frac{B_k}{k!} f^{(k-1)}(m)$$
$$\approx \int_m^\infty f(x)dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m))$$

3.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k), \quad c(0,0) = 1$$
$$\sum_{k=0}^n c(n,k)x^k = x(x+1)\dots(x+n-1)$$

$$c(8,k) = 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$$
$$c(n,2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \dots$$

3.3.3 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j :s s.t. $\pi(j) > \pi(j+1)$, $k+1$ j :s s.t. $\pi(j) \geq j$, k j :s s.t. $\pi(j) > j$.

$$E(n,k) = (n-k)E(n-1,k-1) + (k+1)E(n-1,k)$$

$$E(n,0) = E(n,n-1) = 1$$

$$E(n,k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

3.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n,k) = S(n-1,k-1) + kS(n-1,k)$$

$$S(n,1) = S(n,n) = 1$$

$$S(n,k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

3.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \dots$. For p prime,

$$B(p^m + n) \equiv mB(n) + B(n+1) \pmod{p}$$

3.3.6 Labeled unrooted trees

on n vertices: n^{n-2}
on k existing trees of size n_i : $n_1 n_2 \dots n_k n^{k-2}$
with degrees d_i : $(n-2)! / ((d_1-1)! \dots (d_n-1)!)$

3.3.7 Catalan numbers

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1} = \frac{(2n)!}{(n+1)!n!}$$

$$C_0 = 1, \quad C_{n+1} = \frac{2(2n+1)}{n+2} C_n, \quad C_{n+1} = \sum C_i C_{n-i}$$

$$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \dots$$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n+1$ leaves (0 or 2 children).
- ordered trees with $n+1$ vertices.
- ways a convex polygon with $n+2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

Data Structures (4)

OrderedSet.hpp

Description: Ordered Set

"/template/Header.hpp", <bits/extc++.h> 32a919, 15 lines

using namespace __gnu_pbds;

template <class T>

using ordered_set = tree<T, null_type, less<T>, rb_tree_tag, tree_order_statistics_node_update>;

// can be change to less_equal / greater

void usage() {

ordered_set<int> st, st_2;

st.insert(2);

st.insert(1);

cout << st.order_of_key(2);

cout << *st.find_by_order(1);

st.erase(st.find_by_order(st.order_of_key(5)));

st.join(st_2); // merge

}

SparseTable.hpp

Description: Sparse Table for finding static range queries of monoid operations (in this case min).

Memory: O(N log N)

Time: O(1) query, O(N log N) construction.

eedab3, 18 lines

struct SparseTable{

int n;

vector<vector<int>>> t;

SparseTable() {}

SparseTable(const vector<int> &a) {

n=(int)a.size();

int lg=31-__builtin_clz(n);

t.assign(lg+1,vector<int>(n,le9));

t[0]=a;

for(int i=0;i<lg;i++){

for(int j=0;j+(2<<i)<=n;j++){

t[i+1][j]=min(t[i][j],t[i][j+(1<<i)]);

}

int query(int l,int r){

int lg=31-__builtin_clz(r-l+1);

return min(t[lg][l], t[lg][r-(1<<lg)+1]);

}

}

}

FenwickTree.hpp

Description: Fenwick / Binary Indexed Tree

Memory: O(N)

Time: query and update O(log N)

069d13, 28 lines

struct fenwick {

int n;

vector<int> t;

fenwick (int n=0) {

n=_n;

t.assign(n+1, T{});

}

void update(int x, int v) {

for (int i = x; i <= n; i+=i&-i) t[i] = t[i]+v;

}

void update(int l, int r, int v) {

update(l, v); update(r+1, -v);

}

int query(int x) {

int res = 0;

for (int i = x; i >= 1; i-=i&-i) res = res+t[i];

return res;

}

int query(int l, int r) { return query(r) - query(l-1); }

//find the first index that sums to >= k

int find(const T &k) {

int x = 0;

int cur = 0;

for (int i = 1<<(31-__builtin_clz(n)); i > 0; i>>=1)

if (x+i <= n && cur+t[x+i] < k) x+=i, cur+=t[x];

return x;

}

};

2DFenwickTree.hpp

Description: 2D Fenwick Tree. Requires that elements to be updated is known in advance. Note that X is not compressed.

Memory: O(N log N) (i guess)

Time: query and update O(log^2 N)

8f7a1f, 28 lines

struct TD_fenwick {

int n;

vector<vector<int>>> vals, t;

TD_fenwick(int n, vector<pii> &v): n(n), vals(n+1), t(n+1) {

for (int i = 1; i <= n; ++i) vals[i].push_back(0);

sort(all(v), [&](pii a, pii b){ return a.second < b.second;

});

for (auto [x, y] : v) for (; x <= n; x += x&-x) if (vals[x].back() != y) vals[x].push_back(y);

for (int i = 1; i <= n; ++i) t[i].resize(vals[i].size()+3);

}

inline int cp(int i, int x) {

return upper_bound(all(vals[i]), x) - vals[i].begin();

}

void update(int x, int y, int val) {

for (int i = x; i <= n; i += i&-i) {

for (int j = cp(i, y); j < t[i].size(); j += j&-j) t[i][j] += val;

}

}

int query(int x, int y) {

int res = 0;

for (int i = x; i >= 1; i -= i&-i) {

for (int j = cp(i, y); j >= 1; j -= j&-j) res += t[i][j];

}

return res;

}

int query(int x1, int y1, int x2, int y2) {

return query(x2, y2) - query(x2, y1-1) - query(x1, y2) + query(x1, y1-1);

}

```
    }
};

SegmentTree.hpp
Description: Segment Tree
9b83b9, 82 lines

template<class Node>
struct SegTree {
    int n;
    vector<Node> t;
    SegTree(){};
    SegTree(int n, Node v=Node()) {init(n, v);}
    template<class T>
    SegTree(const vector<T> &a) {init(a);}
    void init(int n, Node v=Node()) {init(vector<Node>(n, v));}
    template<class T>
    void init(const vector<T> &a) {
        n=sz(a);
        t.assign(4<<(31-__builtin_clz(n)),Node());
        function<void(int, int, int)> build=[&](int l, int r, int i
        ) {
            if (l==r) return void(t[i]=a[l-1]);
            int m = (l+r)/2;
            build(l, m, 2*i);
            build(m+1, r, 2*i+1);
            pull(i);
        };
        build(1, n, 1);
    }
    void pull(int i) {t[i] = t[2*i] + t[2*i+1];}
    void modify(int l, int r, int i, int x, const Node &v) {
        if (x<l || x>r) return;
        if (l==r) return void(t[i]=v);
        int m = (l+r)/2;
        if (x<=m) modify(l, m, 2*i, x, v);
        else modify(m+1, r, 2*i+1, x, v);
        pull(i);
    }
    void modify(int x, const Node &v) {modify(1, n, 1, x, v);}
    template<class T>
    void update(int l, int r, int i, int x, const T &v) {
        if (x<l || x>r) return;
        if (l==r) return void(t[i].apply(l, r, v));
        int m = (l+r)/2;
        if (x<=m) update(l, m, 2*i, x, v);
        else update(m+1, r, 2*i+1, x, v);
        pull(i);
    }
    template<class T>
    void update(int x, const T &v) {update(1, n, 1, x, v);}
    Node query(int l, int r, int i, int x, int y) {
        if (y<l || x>r) return Node();
        if (x<=l && r<=y) return t[i];
        int m = (l+r)/2;
        return query(l, m, 2*i, x, y) + query(m+1, r, 2*i+1, x,y);
    }
    Node query(int x, int y) {return query(1, n, 1, x, y);}
    template<class F>
    int findfirst(int l, int r, int i, int x, int y, const F &f)
    {
        if (y<l||r<x||!(f(t[i]))) return -1;
        if (l==r) return l;
        int m = (l+r)/2;
        int ret = findfirst(l, m, 2*i, x, y, f);
        if (ret == -1) ret = findfirst(m+1, r, 2*i+1, x, y, f);
        return ret;
    }
    template<class F>
```

```
    int findfirst(int x, int y, const F &f) {return findfirst(1,
        n, 1, x, y, f);}
    template<class F>
    int findlast(int l, int r, int i, int x, int y, const F &f) {
        if (y<l||r<x||!(f(t[i]))) return -1;
        if (l==r) return l;
        int m = (l+r)/2;
        int ret = findlast(m+1, r, 2*i+1, x, y, f);
        if (ret == -1) ret = findlast(l, m, 2*i, x, y, f);
        return ret;
    }
    template<class F>
    int findlast(int x, int y, const F &f) {return findlast(1, n,
        1, x, y, f);}
};

struct Node {
    ll val;
    Node(ll x=LLONG_MAX):val(x){}
    void apply(int l, int r, int x) {val = x;}
    friend Node operator+(const Node &lhs, const Node &rhs) {
        return Node(min(lhs.val, rhs.val));
    }
};

LazySegmentTree.hpp
Description: Segment Tree with Lazy Propagation
90a572, 81 lines

template<class Node, class Tag>
struct LazySegTree{
    int n;
    vector<Node> t;
    vector<Tag> lz;
    LazySegTree() {}
    LazySegTree(int n, Node v=Node()) {init(n, v);}
    template<class T>
    LazySegTree(const vector<T> &a) {init(a);}
    void init(int n, Node v=Node()) {init(vector<Node>(n, v));}
    template<class T>
    void init(const vector<T> &a) {
        n=sz(a);
        t.assign((4<<(31-__builtin_clz(n)))+1, Node());
        lz.assign((4<<(31-__builtin_clz(n)))+1, Tag());
        function<void(int, int, int)> build=[&](int l, int r, int i
        ) {
            if (l==r) return void(t[i]=a[l-1]);
            int m=(l+r)/2;
            build(l,m,2*i);
            build(m+1,r,2*i+1);
            pull(i);
        };
        build(1,n,1);
    }
    void pull(int i) {
        t[i]=t[2*i]+t[2*i+1];
    }
    void apply(int l,int r,int i,const Tag &v) {
        t[i].apply(l,r,v);
        lz[i].apply(l,r,v);
    }
    void push(int l, int r, int i) {
        int m=(l+r)/2;
        apply(l,m,2*i, lz[i]);
        apply(m+1,r,2*i+1, lz[i]);
        lz[i] = Tag();
    }
    void modify(int l, int r, int i, int x, const Node &v) {
        if (x<l||x>r) return;
        if (l==r) return void(t[i]=v);
```

```
        int m=(l+r)/2;
        push(l,r,i);
        modify(l,m,2*i,x,v);
        modify(m+1,r,2*i+1,x,v);
        pull(i);
    }
    void modify(int x, const Node &v) { modify(1,n,1,x,v); }
    void update(int l, int r, int x, int y, int i, const Tag &v)
    {
        if (y<l||x>r) return;
        if (x<=l&&r<=y) return apply(l, r, i, v);
        int m=(l+r)/2;
        push(l, r, i);
        update(l, m, x, y, 2*i, v);
        update(m+1, r, x, y, 2*i+1, v);
        pull(i);
    }
    void update(int x, int y, const Tag &v) { update(1, n, x, y,
        1, v); }
    void update(int x, const Tag &v) { update(1, n, x, x, 1, v);
        }
    Node query(int l, int r, int x, int y, int i) {
        if (y<l || x>r) return Node();
        if (x<=l&&r<=y) return t[i];
        int m=(l+r)/2;
        push(l, r, i);
        return query(l, m, x, y, 2*i)+query(m+1, r, x, y, 2*i+1);
    }
    Node query(int x, int y) { return query(1, n, x, y, 1); }
    Node query(int x) { return query(1, n, x, x, 1); }
};
struct Tag{
    ll val;
    Tag(ll _val=0):val(_val){}
    void apply(int l, int r, const Tag &v) { val+=v.val; }
};
struct Node{
    ll val;
    Node(ll _val=0):val(_val){}
    void apply(int l, int r, const Tag &v) { val+=v.val; }
    friend Node operator+(const Node &lhs, const Node &rhs) {
        return Node(max(lhs.val, rhs.val));
    }
};
```

```
DynamicSegmentTree.hpp
Description: Dynamic Segment Tree
587ad0, 66 lines

const int MX = 1e5 + 3;

struct Node {
    ll cnt;
    ll sum;
    Node *lc, *rc;
    Node() {
        sum = cnt = 0;
        lc = rc = nullptr;
    }
    void createChild() {
        if (lc == nullptr) lc = new Node();
        if (rc == nullptr) rc = new Node();
    }
};

Node *root = new Node();

void update(ll l, ll r, ll x, ll val, Node *cur) {
    if (l==r) {
        cur -> cnt += val;
```

```

    cur -> sum += x*val;
    return;
}
ll mid = (l+r)>>1;
cur -> createChild();
if (x <= mid) update(l, mid, x, val, cur -> lc);
else update(mid+1, r, x, val, cur -> rc);

cur->sum = cur->lc->sum + cur->rc->sum;
cur->cnt = cur->lc->cnt + cur->rc->cnt;
}

// Query function for the Dynamic Segment Tree
pair<ll, ll> query(ll l, ll r, ll ql, ll qr, Node *cur) {
    if (cur == nullptr || ql > r || qr < l) {
        return {0, 0}; // {count, sum}
    }

    if (ql <= l && r <= qr) {
        return {cur->cnt, cur->sum};
    }

    ll mid = (l + r) >> 1;
    cur->createChild();

    auto left_result = query(l, mid, ql, qr, cur->lc);
    auto right_result = query(mid + 1, r, ql, qr, cur->rc);

    return {left_result.first + right_result.first,
            left_result.second + right_result.second};
}

// Wrapper function for easier usage
pair<ll, ll> query(ll ql, ll qr) {
    return query(0, MX - 1, ql, qr, root);
}

ll search(ll l, ll r, ll lsum, ll rfreq, ll k, Node *cur) {
    if (l==r) return l;
    ll mid = (l+r)>>1;
    cur -> createChild();
    __int128_t cntMid = lsum + (__int128_t)(mid+1)*rfreq + (cur->
        lc->sum) + (__int128_t)(mid+1)*(cur->rc->cnt);
    if (cntMid < (__int128_t)(mid+1)*k) return search(l, mid,
        lsum, rfreq + cur->rc->cnt, k, cur -> lc);
    else return search(mid+1, r, lsum + cur->lc->sum, rfreq, k,
        cur -> rc);
}
}
```

PersistentSegmentTree.hpp

Description: Persistent Segment Tree

\"../template/Header.hpp\"af7fe8, 60 lines

```

const int N = 1<<18;

struct node {
    int lc, rc;
    ll val;
} seg[20*N];
int root[N], cnt, a[N];

void update(int &cur, int l, int r, int x, ll val) {
    cnt++;
    seg[cnt] = seg[cur];
    cur = cnt;

    if (l==r) {
        seg[cur].val = val;
        return;
    }
}
```

PersistentSegmentTree LiChaoTree BinaryTrie

```

int mid = (l+r)/2;
if (x <= mid) {
    update(seg[cur].lc, l, mid, x, val);
}
else {
    update(seg[cur].rc, mid+1, r, x, val);
}
seg[cur].val = seg[seg[cur].lc].val + seg[seg[cur].rc].val;
}

ll query(int cur, int l, int r, int ql, int qr) {
    if (r < ql || l > qr) return 0;
    if (ql <= l && r <= qr) return seg[cur].val;

    int mid = (l+r)/2;
    return query(seg[cur].lc, l, mid, ql, qr) + query(seg[cur].
        rc, mid+1, r, ql, qr);
}

void solve() {
    int n, q; cin >> n >> q;
    for (int i = 1; i <= n; ++i) {
        cin >> a[i];
        update(root[1], 1, n, i, a[i]);
    }

    int sz = 1;

    while (q--) {
        int op; cin >> op;
        if (op == 1) {
            ll k, a, x; cin >> k >> a >> x;
            update(root[k], 1, n, a, x);
        }
        if (op == 2) {
            ll k, a, b; cin >> k >> a >> b;
            cout << query(root[k], 1, n, a, b) << '\n';
        }
        if (op == 3) {
            ll k; cin >> k;
            root[++sz] = root[k];
        }
    }
}
```

LiChaoTree.hpp

Description: Li-Chao Tree.

54de16, 33 lines

```

const int N = 1<<18;

pll seg[N<<1];
ll a[N], qs[N], mqs[N];

ll f(pll p, ll x) {
    return p.st*x + p.nd;
}

void clr(int l, int r, int idx) {
    seg[idx] = {0, -1e18};
    if (l==r) return;
    int m = (l+r)/2;
    clr(l, m, 2*idx);
    clr(m+1, r, 2*idx+1);
}

void update(int l, int r, pll p, int idx) {
    int m = (l+r)/2;
    bool lef = f(p, l) > f(seg[idx], l);
    bool mid = f(p, m) > f(seg[idx], m);
}
```

```

    if (mid) swap(seg[idx], p);
    if (l==r) return;
    if (lef!=mid) update(l, m, p, 2*idx);
    else update(m+1, r, p, 2*idx+1);
}

ll query(int l, int r, int x, int idx) {
    if (l==r) return f(seg[idx], x);
    int m = (l+r)/2;
    if (x <= m) return max(query(l, m, x, 2*idx), f(seg[idx], x
        ));
    else return max(query(m+1, r, x, 2*idx+1), f(seg[idx], x));
}
```

BinaryTrie.hpp

Description: Binary Trie

c96a37, 68 lines

```

template<int BIT>
struct BinaryTrie {
    struct Node {
        array<int, 2> ch;
        int cnt;
        Node() : ch{-1, -1}, cnt(0) {}
    };
    vector<Node> t;
    BinaryTrie() : t{Node{}} {}
    int new_node() {
        t.emplace_back(Node());
        return t.size()-1;
    }
    int size() { return t[0].cnt; }
    bool empty() { return size()==0; }
    void insert(ll val, int k=1) {
        int cur = 0;
        t[cur].cnt += k;
        for (int i = BIT-1; i >= 0; --i) {
            int b = val>>i & 1;
            if (t[cur].ch[b] == -1) t[cur].ch[b] = new_node();
            cur = t[cur].ch[b];
            t[cur].cnt += k;
        }
    }
    void erase(ll val, int k=1) {
        if (count(val) < k) return;
        int cur = 0;
        t[cur].cnt -= k;
        for (int i = BIT-1; i >= 0; --i) {
            int b = val>>i & 1;
            cur = t[cur].ch[b];
            t[cur].cnt -= k;
        }
    }
    void clear() {
        t = {Node};
    }
    ll get_max(ll val) {
        if (empty()) return LLONG_MIN;
        int cur = 0, ans = 0;
        for (int i = BIT-1; i >= 0; --i) {
            int b = val>>i & 1;
            if (t[cur].ch[!b] != -1 && t[t[cur].ch[!b]].cnt > 0) cur
                = t[cur].ch[!b], ans <=< 1, ans++;
            else cur = t[cur].ch[b], ans <=< 1;
        }
        return ans;
    }
    ll get_min(ll val) {
        if (empty()) return LLONG_MAX;
        int cur = 0, ans = 0;
    }
}
```

```

    for (int i = BIT-1; i >= 0; --i) {
        int b = val>>i & 1;
        if (t[cur].ch[b] != -1 && t[t[cur].ch[b]].cnt > 0) cur =
            t[cur].ch[b], ans <= 1, ans++;
        else cur = t[cur].ch[!b], ans <= 1;
    }
    return ans;
}
int count(ll val) {
    int cur = 0;
    for (int i = BIT-1; i >= 0; --i) {
        int b = val>>i & 1;
        if (t[cur].ch[b] == -1) return false;
        cur = t[cur].ch[b];
    }
    return t[cur].cnt;
}
};

```

Mo.hpp

Description: Answer interval or tree path queries by finding an approximate TSP through the queries, and moving from one query to the next by adding/removing points at the ends. If values are on tree edges, change step to add/remove the edge (a, c) and remove the initial add call (but keep in).

Time: $\mathcal{O}(N\sqrt{Q})$

```

void add(int ind, int end) { ... } // add a[ind] (end = 0 or 1)
void del(int ind, int end) { ... } // remove a[ind]
int calc() { ... } // compute current answer

```

```

vi mo(vector<pii> Q) {
    int L = 0, R = 0, blk = 350; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s;
    #define K(x) pii(x.first/blk, x.second ^ -(x.first/blk & 1))
    iota(all(s), 0);
    sort(all(s), [&](int s, int t){ return K(Q[s]) < K(Q[t]); });
    for (int qi : s) {
        pii q = Q[qi];
        while (L > q.first) add(--L, 0);
        while (R < q.second) add(R++, 1);
        while (L < q.first) del(L++, 0);
        while (R > q.second) del(--R, 1);
        res[qi] = calc();
    }
    return res;
}

```

```

vi moTree(vector<array<int, 2>> Q, vector<vi>& ed, int root=0) {
    int N = sz(ed), pos[2] = {}, blk = 350; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s, I(N), L(N), R(N), in(N), par(N);
    add(0, 0), in[0] = 1;
    auto dfs = [&](int x, int p, int dep, auto& f) -> void {
        par[x] = p;
        L[x] = N;
        if (dep) I[x] = N++;
        for (int y : ed[x]) if (y != p) f(y, x, !dep, f);
        if (!dep) I[x] = N++;
        R[x] = N;
    };
    dfs(root, -1, 0, dfs);
    #define K(x) pii(I[x[0]] / blk, I[x[1]] ^ -(I[x[0]] / blk & 1))
    iota(all(s), 0);
    sort(all(s), [&](int s, int t){ return K(Q[s]) < K(Q[t]); });
    for (int qi : s) rep(end, 0, 2) {
        int &a = pos[end], b = Q[qi][end], i = 0;
        #define step(c) { if (in[c]) { del(a, end); in[a] = 0; } \
            else { add(c, end); in[c] = 1; } a = c; }
        while (! (L[b] <= L[a] && R[a] <= R[b]))
            I[i++] = b, b = par[b];
    }
}

```

```

while (a != b) step(par[a]);
while (i--) step(I[i]);
if (end) res[qi] = calc();
}
return res;
}

```

StaticTopTree.hpp

Description: Static Top Tree.

<bits/stdc++.h> 0c26f3, 89 lines

```

using namespace std;

using ll=long long;
const ll N=2e5+5,M=998244353;
vector<ll> g[N];
vector<ll> path[N];
ll par[N],a[N],sz[N],hv[N],top[N],bot[N];

struct node{
    ll m,c,k;
    node *par,*lc,*rc;
    bool s;
    node () : m(0), c(0), k(0), par(NULL), lc(NULL), rc(NULL), s(0) {}
    node (bool s) : m(0), c(0), k(0), par(NULL), lc(NULL), rc(NULL), s(s) {}
    node (bool s,ll k) : m(0), c(0), k(1), par(NULL), lc(NULL), rc(NULL), s(s) {}
    node (bool s,ll m,ll c) : m(m), c(c), k(0), par(NULL), lc(NULL), rc(NULL), s(s) {}
};

node *pt[N],*com[N],*rake[N];

node* build_comp (int l,int r,int nt,int nb,node* np){
    if (l==r){
        int nn=path[nb][1];
        pt[nn]=new node(l,rake[nn]->k,a[nn]);
        rake[nn]->par=pt[nn];
        pt[nn]->par=np;
        pt[nn]->lc=rake[nn];
        return pt[nn];
    }
    node *re=new node(1);
    int mid=(l+r)/2;
    re->lc=build_comp(l,mid,nt,nb,re);
    re->rc=build_comp(mid+1,r,nt,nb,re);
    re->c=(re->rc->c+((re->lc->c*re->rc->m)%M))%M;
    re->m=(re->lc->m*re->rc->m)%M;
    re->par=np;
    return re;
}

node* build_rake(int l,int r,int nr,node* np){
    node *re=new node(0);
    if (l>r){
        re->k=1;
    }
    else if (l==r){
        int nn=g[nr][1];
        if (nn==hv[nr]) re->k=1;
        else{
            com[nn]=build_comp(0,path[bot[nn]].size()-1,nn,bot[nn],re);
            // cout << " comp " << nn << " : " << com[nn]->m <<
                ' ' << com[nn]->c << '\n';
            re->lc=com[nn];
            re->k=com[nn]->c;
        }
    }
}

```

```

}
else{
    int mid=(l+r)/2;
    re->lc=build_rake(l,mid,nr,re);
    re->rc=build_rake(mid+1,r,nr,re);
    re->k=(re->lc->k*re->rc->k)%M;
}
re->par=np;
return re;
}

void dfs (int nn){
    sz[nn]=1;
    int mx=0;
    for (auto e:g[nn]) {
        dfs(e);
        sz[nn]+=sz[e];
        if (sz[e]>mx){
            mx=sz[e];
            hv[nn]=e;
        }
    }
    if (!hv[nn]){
        path[nn].emplace_back(nn);
        top[nn]=nn;
        bot[nn]=nn;
    }
    else{
        path[bot[hv[nn]]].emplace_back(nn);
        top[bot[hv[nn]]]=nn;
        bot[nn]=bot[hv[nn]];
    }
    rake[nn]=build_rake(0,g[nn].size()-1,nn,NULL);
    // cout << " rake " << nn << " : " << rake[nn]->k << '\n';
}

```

Treap.hpp

Description: Treap

<bits/stdc++.h> 8cfd7e, 105 lines

```

using namespace std;

using ll=long long;
int n,q;
mt19937 rnd;

struct node{
    ll prio,val,sum,mn,mx,cnt=1;
    ll rev=0,lz1=0;
    bool f2=0;
    node *lc=NULL,*rc=NULL;
    node(ll val) : prio(rnd()),val(val),sum(val),mn(val),mx(val) {}
    void up(node* nwv){
        if (!nwv) return;
        cnt+=nwv->cnt;
        sum+=nwv->sum;
        mx=max(mx,nwv->mx);
        mn=min(mn,nwv->mn);
        return;
    }
    void in(){
        cnt=1;
        sum=val;
        mn=val;
        mx=val;
    }
};

void push (node* nn,node* np){

```

```
    if (!nn) return;
    if (np->f2){
        nn->f2=1;
        nn->lz1=np->lz1;
    }
    else nn->lz1+=np->lz1;
    nn->rev^=np->rev;
    return;
}

void uplazy (node* nn){
    if (!nn) return;
    if (nn->rev==1){
        swap(nn->lc,nn->rc);
    }
    if (nn->f2){
        nn->val=0;
        nn->mn=0;
        nn->mx=0;
        nn->sum=0;
    }
    nn->val+=nn->lz1;
    nn->mn+=nn->lz1;
    nn->mx+=nn->lz1;
    nn->sum+=nn->cnt * nn->lz1;
    push(nn->lc,nn);
    push(nn->rc,nn);
    nn->lz1=0;
    nn->f2=0;
    nn->rev=0;
    return;
}

node* mg (node* t1,node* t2){
    uplazy(t1); uplazy(t2);
    if (!t1) return t2;
    if (!t2) return t1;
    if (t2->prio < t1->prio){
        t1->up(t2);
        t1->rc=mg(t1->rc,t2);
        return t1;
    }
    else{
        t2->up(t1);
        t2->lc=mg(t1,t2->lc);
        return t2;
    }
}

void sp (node* nn,node* &l,node* &r,int x,ll idx){
    if (!nn) return;
    uplazy(nn);
    if (idx<=x){
        l=nn;
        node* tg=nn->rc;
        uplazy(nn->lc);
        uplazy(nn->rc);
        nn->rc=NULL; nn->in(); nn->up(nn->lc);
        idx++;
        if (tg && tg->lc) idx+=tg->lc->cnt;
        sp(tg,l->rc,r,x,idx);
        l->up(l->rc);
    }
    else{
        r=nn;
        node* tg=nn->lc;
        uplazy(nn->lc);
        uplazy(nn->rc);
        nn->lc=NULL; nn->in(); nn->up(nn->rc);
    }
}
```

```
        idx--;
        if(tg && tg->rc) idx-=tg->rc->cnt;
        sp(tg,l,r->lc,x,idx);
        r->up(r->lc);
    }
    return;
}
```

Number Theory (5)

ExtendedEuclid.hpp

Description: Extended Euclid algorithm for solving diophantine equation ($ax + by = \gcd(a, b)$).

Time: $\mathcal{O}(\log \max\{a, b\})$

```
"../template/Header.hpp" 229e7c, 13 lines

pair<ll,ll> euclid(ll a,ll b){
    ll x=1,y=0,x1=0,y1=1;
    while(b!=0){
        ll q=a/b;
        x-=q*x1;
        y-=q*y1;
        a-=q*b;
        swap(x,x1);
        swap(y,y1);
        swap(a,b);
    }
    return {x,y};
}
```

euclid.h

Description: Finds two integers x and y , such that $ax + by = \gcd(a, b)$. If you just need gcd, use the built in `__gcd` instead. If a and b are coprime, then x is the inverse of $a \pmod{b}$. $x = x_0 + k * (b/g)$ $y = y_0 - k * (a/g)$

```
ll euclid(ll a, ll b, ll &x, ll &y) {
    if (!b) return x = 1, y = 0, a;
    ll d = euclid(b, a % b, y, x);
    return y -= a/b * x, d;
}
```

CRT.hpp

Description: Chinese Remainder Theorem.

$\text{crt}(a, m, b, n)$ computes x such that $x \equiv a \pmod{m}, x \equiv b \pmod{n}$. If $|a| < m$ and $|b| < n$, x will obey $0 \leq x < \text{lcm}(m, n)$. Assumes $mn < 2^{62}$. If x_0 and y_0 is one of the solutions of $ax + by = g$, then the general solution is $x = x_0 + k * (b / g)$ and $y = y_0 - k * (a / g)$.

Time: $\log(n)$

```
"euclid.h" 04d93a, 7 lines

ll crt(ll a, ll m, ll b, ll n) {
    if (n > m) swap(a, b), swap(m, n);
    ll x, y, g = euclid(m, n, x, y);
    assert((a - b) % g == 0); // else no solution
    x = (b - a) % n * x % n / g * m + a;
    return x < 0 ? x + m*n/g : x;
}
```

phiFunction.hpp

Description: Euler's ϕ function is defined as $\phi(n) := \#$ of positive integers $\leq n$ that are coprime with n . $\phi(1) = 1, p$ prime $\Rightarrow \phi(p^k) = (p - 1)p^{k-1}$, m, n coprime $\Rightarrow \phi(mn) = \phi(m)\phi(n)$. If $n = p_1^{k_1}p_2^{k_2}...p_r^{k_r}$ then $\phi(n) = (p_1 - 1)p_1^{k_1-1}...(p_r - 1)p_r^{k_r-1}$. $\phi(n) = n \cdot \prod_{p|n} (1 - 1/p)$. $\sum_{d|n} \phi(d) = n, \sum_{1 \leq k \leq n, \gcd(k,n)=1} k = n\phi(n)/2, n > 1$

Euler's thm: a, n coprime $\Rightarrow a^{\phi(n)} \equiv 1 \pmod{n}$.

Fermat's little thm: p prime $\Rightarrow a^{p-1} \equiv 1 \pmod{p} \forall a$.

```
"efae90, 10 lines"
```

```
const int LIM = 5000000;
int phi[LIM];

void calculatePhi() {
    for(int i=0; i<LIM; ++i) phi[i] = i & 1 ? i : i / 2;
    for (int i = 3; i < LIM; i += 2)
        if (phi[i] == i)
            for (int j = i; j < LIM; j += i)
                phi[j] -= phi[j] / i;
}
```

FloorSum.hpp

Description: Floor sum function. $f(a, b, c, n) = \sum_{x=0}^n \lfloor \frac{ax+b}{c} \rfloor$ becareful when a,b,c are negative (use custom floor division and mod instead)

Time: $\mathcal{O}(\log a)$

```
d088d2, 7 lines

ll floor_sum(ll a,ll b,ll c,ll n){
    ll res=n*(n+1)/2*(a/c)+(n+1)*(b/c);
    a%=c,b%=c;
    if(a==0)return res;
    ll m=(a*n+b)/c;
    return res+n*m-floor_sum(c,c-b-1,a,m-1);
}
```

5.1 Prime Numbers

MillerRabin.hpp

Description: Deterministic Miller-Rabin primality test. Guaranteed to work for numbers up to $7 \cdot 10^{18}$; for larger numbers, use Python and extend A randomly.

Time: 7 times the complexity of $a^b \pmod{c}$.

```
be7e00, 25 lines
```

```
using ull = uint64_t;

ull modmul(ull a, ull b, ull M) {
    ll ret = a * b - M * ull(1.L / M * a * b);
    return ret + M * (ret < 0) - M * (ret >= (1l)M);
}

ull modpow(ull b, ull e, ull mod) {
    ull ans = 1;
    for (; e; b = modmul(b, b, mod), e /= 2)
        if (e & 1) ans = modmul(ans, b, mod);
    return ans;
}

bool isPrime(ull n) {
    if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
    ull A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022},
            s = __builtin_ctzll(n-1), d = n >> s;
    for (ull a : A) { // ^ count trailing zeroes
        ull p = modpow(a%n, d, n), i = s;
        while (p != 1 && p != n - 1 && a % n && i--)
            p = modmul(p, p, n);
        if (p != n-1 && i != s) return 0;
    }
    return 1;
}
```

LinearSieve.hpp

Description: Prime Number Generator in Linear Time

Time: $\mathcal{O}(N)$

```
"../template/Header.hpp" 194fb1, 15 lines

vi linear_sieve(int n) {
    vi prime, composite(n + 1);
    for(int i=2; i<=n; ++i) {
        if(!composite[i]) {
            prime.emplace_back(i);
        }
        for(int j=0; j<(int) prime.size() && i*prime[j]<=n; ++j) {
```

```
        composite[i * prime[j]] = true;
        if(i % prime[j] == 0) {
            break;
        }
    }
}
return prime;
}
```

FastEratosthenes.hpp

Description: Prime sieve for generating all primes smaller than LIM.
Time: LIM=1e9 ≈ 1.5s

"../template/Header.hpp"	295b58, 33 lines
--------------------------	------------------

```
const int LIM = 1e6;
bitset<LIM> isPrime;
vi eratosthenes() {
    const int S = (int) round(sqrt(LIM)), R = LIM / 2;
    vi pr = {2}, sieve(S + 1);
    pr.reserve(int(LIM/log(LIM) * 1.1));
    vector<pii> cp;
    for(int i=3; i<=S; i+=2) {
        if(!sieve[i]) {
            cp.emplace_back(i, i * i / 2);
            for(int j=i*i; j<=S; j+=2*i) {
                sieve[j] = 1;
            }
        }
    }
    for(int L=1; L<=R; L+=S) {
        array<bool, S> block{};
        for(auto &[p, idx]: cp) {
            for(int i=idx; i<S+L; idx=(i+=p)) {
                block[i - L] = 1;
            }
        }
        for(int i=0; i<min(S, R-L); ++i) {
            if(!block[i]) {
                pr.emplace_back((L + i) * 2 + 1);
            }
        }
    }
    for(int i: pr) {
        isPrime[i] = 1;
    }
    return pr;
}
```

GolbatchConjecture.hpp

Description: Find two prime numbers which sum equals s
Time: O(N log N)

"FastEratosthenes.hpp"	88fb23, 18 lines
------------------------	------------------

```
pair<int, int> goldbatchConjecture(int s, vi pr = {}){
    if (s <= 2 || s % 2 != 0) {
        return make_pair(-1, -1);
    }
    if (pr.size() == 0) {
        pr = eratosthenes();
    }
    for (auto x : pr) {
        if (x > s / 2) {
            break;
        }
        int d = s - x;
        if (binary_search(pr.begin(), pr.end(), d)) {
            return make_pair(min(x, d), max(x, d));
        }
    }
    return make_pair(-1, -1);
}
```

"Euclid.h"	45f202, 32 lines
------------	------------------

```
ll cdiv(ll x, ll y) { return x / y + ((x ^ y) > 0 && x % y); }
```

5.2 Modulo

ModArith.hpp

Description: Statistics on a mod'ed arithmetic sequence.
Time: O(log m)

```
// min (ax + b) % m for 0 <= x <= n
ll minRemainder(ll a, ll b, ll m, ll n) {
    assert(a >= 0 && m > 0 && b >= 0 && n >= 0);
    a %= m, b %= m; n = min(n, m - 1);
    if (a == 0) return b;
    if (b >= a) {
        ll ad = cdiv(m - b, a);
        n -= ad; if (n < 0) return b;
        b += ad * a - m;
    }
    ll q = m / a, m2 = m % a;
    if (m2 == 0) return b;
    if (b / m2 > n / q) return b - n / q * m2;
    n -= b / m2 * q; b %= m2;
    ll y2 = (n * a + b) / m;
    ll x2 = cdiv(m2 * y2 - b, a);
    if (x2 * a - m2 * y2 + b >= m2) --x2;
    return minRemainder(a, b, m2, x2);
}

// min x >= 0 s.t. l <= (ax + b) % m <= r
ll minBetween(ll a, ll b, ll m, ll l, ll r) {
    ll x, y, g = euclid(a, m, x, y);
    if (g > 1)
        return minBetween(a/g, b/g, m/g, l/g+(l%g>b%g), r/g-(r%g<b%g));
    if (l > r) return -1; // no solution
    if ((x % m) < 0) x += m;
    ll b2 = (l - b) * x % m;
    return minRemainder(x, b2 < 0 ? b2 + m : b2, m, r - 1);
}
```

ModGen.hpp

Description: Finds a primitive root modulo p.

"Factor.h", "ModMulLL.h"	ff3110, 8 lines
--------------------------	-----------------

```
mt19937_64 rng(2137);
ll modGen(ll n) {
    map<ll, int> f; factor(n - 1, f); rep:
    ll g = rng() % (n - 1) + 1;
    for (auto [p, _] : f)
        if (modpow(g, (n - 1) / p, n) == 1) goto rep;
    return g;
}
```

ModInverse.hpp

Description: Pre-computation of modular inverses. Assumes LIM ≤ mod and that mod is a prime.

"Factor.h", "ModMulLL.h"	6f684f, 3 lines
--------------------------	-----------------

```
const ll mod = 1000000007, LIM = 200000;
ll* inv = new ll[LIM] - 1; inv[1] = 1;
rep(i,2,LIM) inv[i] = mod - (mod / i) * inv[mod % i] % mod;
```

ModLog.hpp

Description: Returns the smallest $x > 0$ s.t. $a^x = b \pmod m$, or -1 if no such x exists. modLog(a,1,m) can be used to calculate the order of a .
Time: O(√m)

"Factor.h", "ModMulLL.h"	c040b8, 11 lines
--------------------------	------------------

```
ll modLog(ll a, ll b, ll m) {
    ll n = (ll) sqrt(m) + 1, e = 1, f = 1, j = 1;
    unordered_map<ll, ll> A;
```

```
while (j <= n && (e = f = e * a % m) != b % m)
    A[e * b % m] = j++;
if (e == b % m) return j;
if (__gcd(m, e) == __gcd(m, b))
    rep(i,2,n+2) if (A.count(e = e * f % m))
        return n * i - A[e];
return -1;
}
```

ModMulLL.hpp

Description: Calculate $a \cdot b \pmod c$ (or $a^b \pmod c$).
Time: O(1) for modmul, O(log b) for modpow

"Factor.h", "ModMulLL.h"	02ea06, 9 lines
--------------------------	-----------------

```
ll modmul(ll a, ll b, ll M) {
    return (__int128)a * b % M;
}
ll modpow(ll b, ll e, ll mod) {
    ll ans = 1;
    for (; e; b = modmul(b, b, mod), e /= 2)
        if (e & 1) ans = modmul(ans, b, mod);
    return ans;
}
```

ModPow.hpp

"Factor.h", "ModMulLL.h"	b83e45, 8 lines
--------------------------	-----------------

```
const ll mod = 1000000007; // faster if const
```

```
ll modpow(ll b, ll e) {
    ll ans = 1;
    for (; e; b = b * b % mod, e /= 2)
        if (e & 1) ans = ans * b % mod;
    return ans;
}
```

ModSqrt.hpp

Description: Tonelli-Shanks algorithm for modular square roots. Finds x s.t. $x^2 = a \pmod p$ ($-x$ gives the other solution).
Time: O(log² p) worst case, O(log p) for most p

"ModMulLL.h"	b7cab4, 24 lines
--------------	------------------

```
ll sqrt(ll a, ll p) {
    a %= p; if (a < 0) a += p;
    if (a == 0) return 0;
    if (modpow(a, (p-1)/2, p) != 1) return -1; // no solution
    if (p % 4 == 3) return modpow(a, (p+1)/4, p);
    // a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8 == 5
    ll s = p - 1, n = 2;
    int r = 0, m;
    while (s % 2 == 0)
        ++r, s /= 2;
    while (modpow(n, (p - 1) / 2, p) != p - 1) ++n;
    ll x = modpow(a, (s + 1) / 2, p);
    ll b = modpow(a, s, p), g = modpow(n, s, p);
    for (; r = m) {
        ll t = b;
        for (m = 0; m < r && t != 1; ++m)
            t = t * t % p;
        if (m == 0) return x;
        ll gs = modpow(g, 1LL << (r - m - 1), p);
        g = gs * gs % p;
        x = x * gs % p;
        b = b * g % p;
    }
}
```

ModSum.hpp

Description: Tonelli-Shanks algorithm for modular square roots. Finds x s.t. $x^2 = a \pmod p$ ($-x$ gives the other solution).

Time: $\mathcal{O}(\log^2 p)$ worst case, $\mathcal{O}(\log p)$ for most p

"ModModLL.h"b7cab4, 24 lines

```
11 sqrt(1l a, 1l p) {
    a %= p; if (a < 0) a += p;
    if (a == 0) return 0;
    if (modpow(a, (p-1)/2, p) != 1) return -1; // no solution
    if (p % 4 == 3) return modpow(a, (p+1)/4, p);
    // a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8 == 5
    1l s = p - 1, n = 2;
    int r = 0, m;
    while (s % 2 == 0)
        ++r, s /= 2;
    while (modpow(n, (p - 1) / 2, p) != p - 1) ++n;
    1l x = modpow(a, (s + 1) / 2, p);
    1l b = modpow(a, s, p), g = modpow(n, s, p);
    for (; r = m) {
        1l t = b;
        for (m = 0; m < r && t != 1; ++m)
            t = t * t % p;
        if (m == 0) return x;
        1l gs = modpow(g, 1LL << (r - m - 1), p);
        g = gs * gs % p;
        x = x * gs % p;
        b = b * g % p;
    }
}
```

ModArithmetic.hpp

Description: Operators for modular arithmetic. You need to set mod to some number first and then you can use the structure.

"euclid.h"35bfea, 18 lines

```
const 1l mod = 17; // change to something else
struct Mod {
    1l x;
    Mod(1l xx) : x(xx) {}
    Mod operator+(Mod b) { return Mod((x + b.x) % mod); }
    Mod operator-(Mod b) { return Mod((x - b.x + mod) % mod); }
    Mod operator*(Mod b) { return Mod((x * b.x) % mod); }
    Mod operator/(Mod b) { return *this * invert(b); }
    Mod invert(Mod a) {
        1l x, y, g = euclid(a.x, mod, x, y);
        assert(g == 1); return Mod((x + mod) % mod);
    }
    Mod operator^(1l e) {
        if (!e) return Mod(1);
        Mod r = *this ^ (e / 2); r = r * r;
        return e&1 ? *this * r : r;
    }
};
```

Graph (6)

SCC.hpp

Description: Strongly Connected Component.

ed52c6, 20 lines

```
vi adj[MX], rev[MX];
int comp[MX], cnt;
bool vis[MX];
stack<int> order;

void dfs(int i) {
    if (vis[i]) return;
    vis[i] = 1;
    trav(j, adj[i]) dfs(j);
    order.push(i);
}
```

void dfs2(int i, int c) {

if (comp[i]) return;

comp[i] = c;

for (auto j : rev[i]) dfs2(j, c);

}

rep(i, 1, M) dfs(i);

while (!order.empty()) dfs2(order.top(), ++cnt), order.pop();

ArticulationPoint.hpp

Description: Articulation Point

31c52f, 31 lines

```
// adj[u] = adjacent nodes of u
// ap = AP = articulation points
// p = parent
// disc[u] = discovery time of u
// low[u] = 'low' node of u

int dfsAP(int u, int p) {
    int children = 0;
    low[u] = disc[u] = ++Time;
    for (int& v : adj[u]) {
        if (v == p) continue; // we don't want to go back through
                               // the same path.
                               // if we go back is because we found
                               // another way back
        if (!disc[v]) { // if V has not been discovered before
            children++;
            dfsAP(v, u); // recursive DFS call
            if (disc[u] <= low[v]) // condition #1
                ap[u] = 1;
            low[u] = min(low[u], low[v]); // low[v] might be an
                                         // ancestor of u
        } else // if v was already discovered means that we found
                // an ancestor
            low[u] = min(low[u], disc[v]); // finds the ancestor with
                                         // the least discovery time
    }
    return children;
}

void AP() {
    ap = low = disc = vector<int>(adj.size());
    Time = 0;
    for (int u = 0; u < adj.size(); u++)
        if (!disc[u])
            ap[u] = dfsAP(u, u) > 1; // condition #2
}
```

Bridge.hpp

Description: Bridge

f7f699, 17 lines

```
vi adj[MX];
int disc[MX], low[MX], s[MX], t;
1l ans;
map<pii, pii> bridge;

void find_bridge(int u, int prt, int n) {
    disc[u] = low[u] = ++t, s[u] = 1;
    for (auto v : adj[u]) {
        if (v == prt) continue;
        if (!disc[v]) {
            find_bridge(v, u, n);
            if (disc[u] < low[v]) ckmax(ans, (1l)s[u]*(n-s[v]));
            s[u] += s[v];
        }
        ckmin(low[u], low[v]);
    }
}
```

Hierholzer.hpp

Description: Hierholzer Algorithm for finding Eulerian Circuit.

92be7f, 9 lines

```
void dfs(int u) {
    while (!adj[u].empty()) {
        auto [v, idx] = adj[u].back(); adj[u].pop_back();
        if (walk[idx]) continue;
        walk[idx] = 1;
        dfs(v);
        ans[++c] = idx;
    }
}
```

6.1 Matching

HopcroftKarp.hpp

Description: Fast bipartite matching algorithm.

Time: $\mathcal{O}(E\sqrt{V})$

"../template/Header.hpp"Obd56f, 52 lines

```
struct HopcroftKarp{
    int n,m;
    vi l,r,lv,ptr;
    vector<vi> adj;
    HopcroftKarp(){}
    HopcroftKarp(int _n,int _m){init(_n,_m);}
    void init(int _n,int _m){
        n=_n,m=_m;
        adj.assign(n+m,vi{});
    }
    void addEdge(int u,int v){
        adj[u].emplace_back(v+n);
    }
    void bfs(){
        lv=vi(n,-1);
        queue<int> q;
        for(int i=0;i<n;i++)if(l[i]==-1){
            lv[i]=0;
            q.emplace(i);
        }
        while(!q.empty()){
            int u=q.front();
            q.pop();
            for(int v:adj[u])if(r[v]!=-1&&lv[r[v]]==-1){
                lv[r[v]]=lv[u]+1;
                q.emplace(r[v]);
            }
        }
    }
    bool dfs(int u){
        for(int &i=ptr[u];i<sz(adj[u]);i++){
            int v=adj[u][i];
            if(r[v]==-1||(lv[r[v]]==lv[u]+1&&dfs(r[v]))){
                l[u]=v,r[v]=u;
                return true;
            }
        }
        return false;
    }
    int maxMatching(){
        int match=0,cnt=0;
        l=r=vi(n+m,-1);
        do{
            ptr=vi(n);
            bfs();
            cnt=0;
            for(int i=0;i<n;i++)if(l[i]==-1&&dfs(i))cnt++;
            match+=cnt;
        }while(cnt);
        return match;
    }
};
```

```
    }
};

Kuhn.hpp
Description: Kuhn Algorithm to find maximum bipartite matching or find augmenting path in bipartite graph.
Time:  $O(VE)$ 
"../template/Header.hpp" 4b91e8, 27 lines
```

```
vi adj[1010], match(1010, -1);
vector<bool> visited(1010, false);
bool try_match(int u) {
    if(visited[u]) {
        return false;
    }
    visited[u] = true;
    for(auto x: adj[u]) {
        if(match[x] == -1 || try_match(match[x])) {
            match[x] = u;
            return true;
        }
    }
    return false;
}

int max_matching() {
    for(int u=0; u<1010; ++u) {
        visited = vector<bool> (1010, false);
        try_match(u);
    }
    int cnt = 0;
    for(int u=0; u<1010; ++u) {
        cnt += (match[u] != -1);
    }
    return cnt;
}
```

```
WeightedMatching.hpp
Description: Given a weighted bipartite graph, matches every node on the left with a node on the right such that no nodes are in two matchings and the sum of the edge weights is minimal. Takes cost[N][M], where cost[i][j] = cost for L[i] to be matched with R[j] and returns (min cost, match), where L[i] is matched with R[match[i]]. Negate costs for max cost. Requires  $N \leq M$ .
Time:  $O(N^2M)$ 
2540b8, 34 lines
```

```
pair<ll, vector<int>>> hungarian(const vector<vector<ll>>> &a) {
    if (a.empty()) return {0, {}};
    int n = a.size() + 1, m = a[0].size() + 1;
    vector<ll> u(n), v(m);
    vector<int> p(m), ans(n - 1);
    for(int i=1;i<n;i++) {
        p[0] = i;
        int j0 = 0; // add "dummy" worker 0
        vector<ll> dist(m, LLONG_MAX);
        vector<int> pre(m, -1);
        vector<bool> done(m + 1);
        do { // dijkstra
            done[j0] = true;
            int i0 = p[j0], j1;
            ll delta = LLONG_MAX;
            for(int j=1;j<m;j++) if (!done[j]) {
                auto cur = a[i0 - 1][j - 1] - u[i0] - v[j];
                if (cur < dist[j]) dist[j] = cur, pre[j] = j0;
                if (dist[j] < delta) delta = dist[j], j1 = j;
            }
            for(int j=0;j<m;j++) {
                if (done[j]) u[p[j]] += delta, v[j] -= delta;
                else dist[j] -= delta;
            }
            j0 = j1;
        }
```

```
    } while (p[j0]);
    while (j0) { // update alternating path
        int j1 = pre[j0];
        p[j0] = p[j1], j0 = j1;
    }
}
for(int j=1;j<m;j++) if (p[j]) ans[p[j] - 1] = j - 1;
return {-v[0], ans}; // min cost
}
```

6.2 Network Flow

```
Dinic.hpp
Description: Dinic's Algorithm for finding the maximum flow.
Time:  $O(VE \log U)$  where U is the maximum flow.
2b9ab1, 88 lines
```

```
template<class T,bool directed=true,bool scaling=true>
struct Dinic{
    static constexpr T INF=numeric_limits<T>::max()/2;
    struct Edge{
        int to;
        T flow,cap;
        Edge(int _to,T _cap):to(_to),flow(0),cap(_cap){}
        T remain(){return cap-flow;}
    };
    int n,s,t;
    T U;
    vector<Edge> e;
    vector<vector<int>>> g;
    vector<int> ptr,lv;
    bool calculated;
    T max_flow;
    Dinic(){
        Dinic(int n,int s,int t){init(n,s,t);}
        void init(int _n,int _s,int _t){
            n=_n,s=_s,t=_t;
            U=0;
            e.clear();
            g.assign(n,{});
            calculated=false;
        }
        void add_edge(int from,int to,T cap){
            assert(0<=from&&from<n&&0<=to&&to<n);
            g[from].emplace_back(e.size());
            e.emplace_back(to,cap);
            g[to].emplace_back(e.size());
            e.emplace_back(from,directed?0:cap);
            U=max(U,cap);
        }
        bool bfs(T scale){
            lv.assign(n,-1);
            vector<int> q{s};
            lv[s]=0;
            for(int i=0;i<(int)q.size();i++){
                int u=q[i];
                for(int j:g[u]){
                    int v=e[j].to;
                    if(lv[v]==-1&&e[j].remain()>=scale){
                        q.emplace_back(v);
                        lv[v]=lv[u]+1;
                    }
                }
            }
            return lv[t]!=-1;
        }
        T dfs(int u,int t,T f){
            if(u==t||f==0) return f;
            for(int &i=ptr[u];i<(int)g[u].size();i++){
                int j=g[u][i];
                int v=e[j].to;
                if(lv[v]==lv[u]+1){
                    T res=dfs(v,t,min(f,e[j].remain()));
                    if(res>0){
                        e[j].flow+=res;
                        e[j^1].flow-=res;
                        return res;
                    }
                }
            }
            return 0;
        }
        T flow(){
            if(calculated)return max_flow;
            calculated=true;
            max_flow=0;
            for(T scale=scaling?1LL<<(63-__builtin_clz1l(U)):1LL;
                scale>0;scale>>=1){
                while(bfs(scale)){
                    ptr.assign(n,0);
                    while(true){
                        T f=dfs(s,t,INF);
                        if(f==0)break;
                        max_flow+=f;
                    }
                }
            }
            return max_flow;
        }
        pair<T,vector<int>>> cut(){
            flow();
            vector<int> res(n);
            for(int i=0;i<n;i++) res[i]=(lv[i]==-1);
            return {max_flow,res};
        }
    };
};

MinCostFlow.hpp
Description: minimum-cost flow algorithm.
Time:  $O(FE \log V)$  where F is max flow.
8ea1d2, 83 lines
```

```
    if(lv[v]==lv[u]+1){
        T res=dfs(v,t,min(f,e[j].remain()));
        if(res>0){
            e[j].flow+=res;
            e[j^1].flow-=res;
            return res;
        }
    }
    return 0;
}

T flow(){
    if(calculated)return max_flow;
    calculated=true;
    max_flow=0;
    for(T scale=scaling?1LL<<(63-__builtin_clz1l(U)):1LL;
        scale>0;scale>>=1){
        while(bfs(scale)){
            ptr.assign(n,0);
            while(true){
                T f=dfs(s,t,INF);
                if(f==0)break;
                max_flow+=f;
            }
        }
    }
    return max_flow;
}

pair<T,vector<int>>> cut(){
    flow();
    vector<int> res(n);
    for(int i=0;i<n;i++) res[i]=(lv[i]==-1);
    return {max_flow,res};
}
};

MinCostFlow.hpp
Description: minimum-cost flow algorithm.
Time:  $O(FE \log V)$  where F is max flow.
8ea1d2, 83 lines
```

```
template<class F,class C>
struct MinCostFlow{
    struct Edge{
        int to;
        F flow,cap;
        C cost;
        Edge(int _to,F _cap,C _cost):to(_to),flow(0),cap(_cap),
            cost(_cost){}
        F getcap(){
            return cap-flow;
        }
    };
    int n;
    vector<Edge> e;
    vector<vi> adj;
    vector<C> pot,dist;
    vi pre;
    bool neg;
    const F FINF=numeric_limits<F>::max()/2;
    const C CINF=numeric_limits<C>::max()/2;
    MinCostFlow(){
        MinCostFlow(int _n){
            init(_n);
        }
        void init(int _n){
            n=_n;
            e.clear();
            adj.assign(n,{});
            neg=false;
        }
    };
};
```

```

}
void addEdge(int u,int v,F cap,C cost){
    adj[u].emplace_back(sz(e));
    e.emplace_back(v,cap,cost);
    adj[v].emplace_back(sz(e));
    e.emplace_back(u,0,-cost);
    if(cost<0)neg=true;
}
bool dijkstra(int s,int t){
    using P = pair<C,int>;
    dist.assign(n,CINF);
    pre.assign(n,-1);
    priority_queue<P,vector<P>,greater<P>> pq;
    dist[s]=0;
    pq.emplace(0,s);
    while(!pq.empty()){
        auto [d,u]=pq.top();
        pq.pop();
        if(dist[u]<d)continue;
        for(int i:adj[u]){
            int v=e[i].to;
            C ndist=d+pot[u]-pot[v]+e[i].cost;
            if(e[i].getcap()>0&&dist[v]>ndist){
                pre[v]=i;
                dist[v]=ndist;
                pq.emplace(ndist,v);
            }
        }
    }
    return dist[t]<CINF;
}
pair<F,C> flow(int s,int t){
    F flow=0;
    C cost=0;
    pot.assign(n,0);
    if(neg)for(int t=0;t<n;t++)for(int i=0;i<sz(e);i++)if(e[i].getcap()>0){
        int u=e[i^1].to,v=e[i].to;
        pot[v]=min(pot[v],pot[u]+e[i].cost);
    } // Bellman-Ford
    while(dijkstra(s,t)){
        for(int i=0;i<n;i++)pot[i]+=dist[i];
        F aug=FINF;
        for(int u=t;u!=s;u=e[pre[u]^1].to){
            aug=min(aug,e[pre[u]].getcap());
        } // find bottleneck
        for(int u=t;u!=s;u=e[pre[u]^1].to){
            e[pre[u]].flow+=aug;
            e[pre[u]^1].flow-=aug;
        } // push flow
        flow+=aug;
        cost+=aug*pot[t];
    }
    return {flow,cost};
}
};

```

Tree (7)

LCA.hpp

Description: LCA

523b28, 27 lines

```

vi adj[MX];
int up[MX][L+1], lvl[MX];

void dfs(int u, int prt) {
    lvl[u] = lvl[prt] + 1;
    up[u][0] = prt;

```

```

    rep(i, 1, L) up[u][i] = up[up[u][i-1]][i-1];
    for (auto v : adj[u]) {
        if (v == prt) continue;
        dfs(v, u);
    }
}

```

```

int ancestor(int u, int a) {
    rep(i, 0, L) if (a&(1<<i)) u = up[u][i];
    return u;
}

```

```

int lca(int a, int b) {
    if (lvl[a] < lvl[b]) swap(a, b);
    a = ancestor(a, lvl[a] - lvl[b]);
    if (a==b) return a;
    repd(i, 0, L)
        if (up[a][i] != up[b][i])
            a = up[a][i], b = up[b][i];
    return up[a][0];
}

```

CartesianTree.hpp

Description: Definite integral using Simpson's formula.

ad3b5d, 40 lines

```

template<class T,bool IS_MIN>
struct CartesianTree{
    int n;
    vector<T> &a;
    vector<pair<int,int>> range;
    vector<int> lch,rch,par;
    int root;

    CartesianTree(vector<T> &a):n((int)_a.size()),a(_a){
        range.assign(n,{-1,-1});
        lch=rch=par=vector<int>(n,-1);
        if(n==1){
            range[0]={0,1};
            root=0;
            return;
        }
        auto cmp=[&](int i,int j)->bool {
            if(IS_MIN)a[i]<a[j]||(a[i]==a[j]&&i<j);
            return a[i]>a[j]||(a[i]==a[j]&&i<j);
        };
        vector<int> st;
        for(int i=0;i<n;i++){
            while(!st.empty()&&cmp(i,st.back())){
                lch[i]=st.back();
                st.pop_back();
            }
            range[i].first=(st.empty()?-1:st.back()+1);
        }
        for(int i=n-1;i>=0;i--){
            while(!st.empty()&&cmp(i,st.back())){
                rch[i]=st.back();
                st.pop_back();
            }
            range[i].second=(st.empty()?n:st.back()-1);
        }
    }
};

```

VirtualTree.hpp

Description: Virtual Tree of some problem.

<bits/stdc++.h>

575619, 114 lines

```

using namespace std;

const int N = 3e5 + 9;

vector<int> g[N];
int par[N][20], dep[N], sz[N], st[N], en[N], T;
void dfs(int u, int pre) {
    par[u][0] = pre;
    dep[u] = dep[pre] + 1;
    sz[u] = 1;
    st[u] = ++T;
    for (int i = 1; i <= 18; i++) par[u][i] = par[par[u][i-1]][i-1];
    for (auto v : g[u]) {
        if (v == pre) continue;
        dfs(v, u);
        sz[u] += sz[v];
    }
    en[u] = T;
}
int lca(int u, int v) {
    if (dep[u] < dep[v]) swap(u, v);
    for (int k = 18; k >= 0; k--) if (dep[par[u][k]] >= dep[v]) u = par[u][k];
    if (u == v) return u;
    for (int k = 18; k >= 0; k--) if (par[u][k] != par[v][k]) u = par[u][k], v = par[v][k];
    return par[u][0];
}
int kth(int u, int k) {
    for (int i = 0; i <= 18; i++) if (k & (1 << i)) u = par[u][i];
    return u;
}
int dist(int u, int v) {
    int lc = lca(u, v);
    return dep[u] + dep[v] - 2 * dep[lc];
}
int isanc(int u, int v) {
    return (st[u] <= st[v]) && (en[v] <= en[u]);
}
vector<int> t[N];
// given specific nodes, construct a compressed directed tree
// with these vertices(if needed some other nodes included)
// returns the nodes of the tree
// nodes.front() is the root
// t[] is the specific tree
vector<int> buildtree(vector<int> v) {
    // sort by entry time
    sort(v.begin(), v.end(), [](int x, int y) {
        return st[x] < st[y];
    });
    // finding all the ancestors, there are few of them
    int s = v.size();
    for (int i = 0; i < s - 1; i++) {
        int lc = lca(v[i], v[i+1]);
        v.push_back(lc);
    }
    // removing duplicated nodes
    sort(v.begin(), v.end());
    v.erase(unique(v.begin(), v.end()), v.end());
    // again sort by entry time
    sort(v.begin(), v.end(), [](int x, int y) {
        return st[x] < st[y];
    });
    stack<int> st;

```

```
st.push(v[0]);
for (int i = 1; i < v.size(); i++) {
    while (!isanc(st.top(), v[i])) st.pop();
    t[st.top()].push_back(v[i]);
    st.push(v[i]);
}
return v;
}

int ans;
int imp[N];
int yo(int u) {
    vector<int> nw;
    for (auto v : t[u]) nw.push_back(yo(v));
    if (imp[u]) {
        for (auto x : nw) if (x) ans++;
        return 1;
    } else {
        int cnt = 0;
        for (auto x : nw) cnt += x > 0;
        if (cnt > 1) {
            ans++;
            return 0;
        }
        return cnt;
    }
}

int32_t main() {
    ios_base::sync_with_stdio(0);
    cin.tie(0);
    int i, j, k, n, m, q, u, v;
    cin >> n;
    for (i = 1; i < n; i++) cin >> u >> v, g[u].push_back(v), g[v]
        ].push_back(u);
    dfs(1, 0);
    cin >> q;
    while (q--) {
        cin >> k;
        vector<int> v;
        for (i = 0; i < k; i++) cin >> m, v.push_back(m), imp[m] =
            1;
        int fl = 1;
        for (auto x : v) if (imp[par[x][0]]) fl = 0;
        ans = 0;
        vector<int> nodes;
        if (fl) nodes = buildtree(v);
        if (fl) yo(nodes.front());
        if (!fl) ans = -1;
        cout << ans << '\n';
        // clear the tree
        for (auto x : nodes) t[x].clear();
        for (auto x : v) imp[x] = 0;
    }
    return 0;
}

// https://codeforces.com/contest/613/problem/D
```

HLD.hpp

Description: HLD

../template/Header.hpp

cf6882, 45 lines

vector<vi> adj;
vector<int> sz, lvl, hv, hd, p, disc;
int t;

void dfs(int u, int parent) {
 sz[u] = 1;
 lvl[u] = lvl[parent] + 1;
 p[u] = parent;
 int c_hv=0, c_max=0;
 for(auto v: adj[u]) {

```
if(v == parent) continue;
dfs(v, u);
sz[u] += sz[v];
if(c_max < sz[v]) {
    c_hv = v;
    c_max = sz[v];
}
}
hv[u] = c_hv;
}

void hld(int u, int parent) {
    if(hd[u] == 0) {
        hd[u] = u;
    }
    disc[u] = ++t;
    if(hv[u] != 0) {
        hd[hv[u]] = hd[u];
        hld(hv[u], u);
    }
    for(auto v: adj[u]) {
        if(v == parent || v == hv[u]) {
            continue;
        }
        hld(v, u);
    }
}

int lca(int u, int v) {
    while(hd[u] != hd[v]) {
        if(lvl[hd[u]] > lvl[hd[v]]) swap(u, v);
        v=p[hd[v]];
    }
    return lvl[u] < lvl[v] ? u: v;
}
```

CentroidDecom.hpp

Description: Centroid

../template/Header.hpp

e46d44, 32 lines

vector<vi> adj;
vi sz;
vector<bool> used;

int find_size(int u, int p) {
 sz[u] = 1;
 for(auto v: adj[u]) {
 if(v == p || used[v]) continue;
 sz[u] += find_size(v, u);
 }
 return sz[u];
}

int find_cen(int u, int p, int t) {
 for(auto v: adj[u]) {
 if(v == p || used[v]) continue;
 if(sz[v] * 2 > t) find_cen(v, u, t);
 }
 return u;
}

void decom(int u) {
 u = find_cen(u, 0, find_size(u, 0));
 used[u] = true;
 for(auto v: adj[u]) {
 // dfs do something
 }
 for(auto v: adj[u]) {
 if(used[v]) continue;
 decom(v);
 }
}

RootedTreeIsomorphism.hpp

Description: Rooted Tree Isomorphism Check

7fe119, 29 lines

const int MX = 1e5 + 3;

vi adj[2][MX];
map<vector<int>, int> mp;
int cnt;

int dfs(int i, int u, int prt) {
 vector<int> m;
 for (auto v : adj[i][u]) {
 if (v == prt) continue;
 m.pb(dfs(i, v, u));
 }
 sort(all(m));
 if (!mp.count(m)) mp[m] = ++cnt;
 return mp[m];
}

void solve() {
 int n; cin >> n;
 rep(j, 2) rep(i, 2, n) {
 int u, v; cin >> u >> v;
 adj[j][u].pb(v); adj[j][v].pb(u);
 }
 cout << (dfs(0, 1, 0) == dfs(1, 1, 0) ? "YES" : "NO") << nl;
 //clear previous testcase
 rep(j, 2) rep(i, 1, n) adj[j][i].clear();
 mp.clear();
 cnt = 0;
}

UnrootedTreeIsomorphism.hpp

Description: Unrooted Tree Isomorphism Check

642f54, 45 lines

const int MX = 1e5 + 3;

vi adj[MX][2];
int sz[MX][2], cnt;
vector<int> centroid[2];
map<vector<int>, int> mp;

void find_centroid(int u, int t, int n, int prt=0) {
 bool is_centroid = 1;
 sz[u][t] = 1;
 for (auto v : adj[u][t]) {
 if (v == prt) continue;
 find_centroid(v, t, n, u);
 if (sz[v][t] > n/2) is_centroid = 0;
 sz[u][t] += sz[v][t];
 }
 if (n - sz[u][t] > n/2) is_centroid = 0;
 if (is_centroid) centroid[t].pb(u);
}

int dfs(int u, int t, int prt=0) {
 vector<int> m;
 for (auto v : adj[u][t]) {
 if (v == prt) continue;
 m.pb(dfs(v, t, u));
 }
 sort(all(m));
 if (!mp.count(m)) mp[m] = ++cnt;
 return mp[m];
}

```
void solve() {
    int n; cin >> n;
    rep(j, 2) rep(i, 2, n) {
        int u, v; cin >> u >> v;
        adj[u][j].pb(v), adj[v][j].pb(u);
    }
    rep(j, 2) find_centroid(1, j, n);
    bool iso = 0;
    for (auto c0 : centroid[0]) for (auto c1 : centroid[1]) {
        iso |= dfs(c0, 0) == dfs(c1, 1);
    }
    cout << (iso ? "YES" : "NO") << nl;
    rep(j, 2) rep(i, 1, n) adj[i][j].clear();
    rep(j, 2) centroid[j].clear();
}
```

Strings (8)

KMP.hpp

Description: pi[x] computes the length of the longest prefix of s that ends at x, other than s[0...x] itself (abacaba -> 0010123). Can be used to find all occurrences of a string.

Time: $\mathcal{O}(N)$

d4375c, 16 lines

```
vi pi(const string& s) {
    vi p(sz(s));
    rep(i, 1, sz(s)) {
        int g = p[i-1];
        while (g && s[i] != s[g]) g = p[g-1];
        p[i] = g + (s[i] == s[g]);
    }
    return p;
}

vi match(const string& s, const string& pat) {
    vi p = pi(pat + '\0' + s), res;
    rep(i, sz(p)-sz(s), sz(p))
        if (p[i] == sz(pat)) res.push_back(i - 2 * sz(pat));
    return res;
}
```

ZAlgo.hpp

Description: z[x] computes the length of the longest common prefix of s[i:] and s, except z[0] = 0. (abacaba -> 0010301)

"../template/Header.hpp"

58dc79, 11 lines

```
vector<int> z_algorithm(const string &s){
    int n=(int)s.size();
    vector<int> z(n);
    z[0]=n;
    for(int i=1,l=0,r=1;i<n;i++){
        if(i<r)z[i]=min(r-i,z[i-l]);
        while(i+z[i]<n&&s[z[i]]==s[i+z[i]])z[i]++;
        if(i+z[i]>r)l=i,r=i+z[i];
    }
    return z;
}
```

AhoCorasick.hpp

Description: Aho Corasick Automaton

13041a, 75 lines

```
struct AC {
    struct Node {
        ll val = 0, mark = 0;
        vector<int> ch;
        Node() { ch.assign(27, 0); }
    };
    vector<Node> t;
    vector<int> out, link;
```

```
AC () { push_back(); }
int push_back() {
    t.push_back(Node());
    out.push_back(0);
    link.push_back(0);
    return t.size()-1;
}

int insert(const string s) {
    int idx = 0;
    for (auto c : s) {
        int v = (c==' '?26:c-'a');
        if (!t[idx].ch[v]) t[idx].ch[v] = push_back();
        idx = t[idx].ch[v];
    }
    t[idx].mark = 1;
    return idx;
}

void compute() {
    queue<int> q; q.push(0);
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (int c = 0; c < 27; ++c) {
            int v = t[u].ch[c];
            if (!v) t[u].ch[c] = t[link[u]].ch[c];
            else {
                link[v] = u ? t[link[u]].ch[c] : 0;
                out[v] = (t[link[v]].mark ? link[v] : out[link[v]]);
                q.push(v);
            }
        }
    }

    int advance(int idx, char c) {
        int v = (c==' '?26:c-'a');
        while (idx && !t[idx].ch[v]) idx = link[idx];
        return t[idx].ch[v];
    }
};

string p[MX];
ll idx[MX], po[MX];

signed main() {
    ios_base::sync_with_stdio(0); cin.tie(NULL);
    int n, m, k; cin >> n >> m >> k;
    AC aho;
    string s; getline(cin, s);
    for (int i = 1; i <= n; ++i) getline(cin, p[i]), idx[i] = aho
        .insert(p[i]);
    aho.compute();
    po[0] = 1;
    for (int i = 1; i <= m; ++i) po[i] = (po[i-1] * 173) % MOD;
    for (int i = 1; i <= m; ++i) {
        int u, v; cin >> u >> v;
        aho.t[idx[u]].val = (aho.t[idx[u]].val + po[i]) % MOD;
        aho.t[idx[v]].val = (aho.t[idx[v]].val - po[i]) % MOD;
    }
    getline(cin, s);
    for (int i = 1; i <= k; ++i) {
        string s; getline(cin, s);
        ll val = 0, u = 0;
        for (auto c : s) {
            u = aho.advance(u, c);
            for (int v = u; v; v = aho.out[v]) val = (val + aho.t[v].
                val) % MOD;
        }
        cout << (val ? "no" : "yes") << nl;
    }
}
```

Ukkonen.hpp

Description: Ukkonen's Algorithm for suffix tree construction

643022, 59 lines

```
const int N=1000000, // maximum possible number of nodes in
    suffix tree
    INF=1000000000; // infinity constant
string a; // input string for which the suffix tree is
    being built
int t[N][26], // array of transitions (state, letter)
    l[N], // left ...
    r[N], // ...and right boundaries of the substring of a
    which correspond to incoming edge
    p[N], // parent of the node
    s[N], // suffix link
    tv, // the node of the current suffix (if we're mid-
    edge, the lower node of the edge)
    tp, // position in the string which corresponds to the
    position on the edge (between l[tp] and r[tp],
    inclusive)
    ts, // the number of nodes
    la; // the current character in the string

void ukkadd(int c) { // add character s to the tree
    suff;; // we'll return here after each transition to
    the suffix (and will add character again)
    if (r[tp]<tp) { // check whether we're still within the
        boundaries of the current edge
        // if we're not, find the next edge. If it doesn't
        exist, create a leaf and add it to the tree
        if (t[tp][c]==-1) {t[tp][c]=ts;l[ts]=la;p[ts++]=tp;tv=s
            [tp];tp=r[tp]+1;goto suff;}
        tv=t[tp][c];tp=l[tp];
    } // otherwise just proceed to the next edge
    if (tp==-1 || c==a[tp]-'a')
        tp++; // if the letter on the edge equal c, go down
        that edge
    else {
        // otherwise split the edge in two with middle in node
        ts
        l[ts]=l[tp];r[ts]=tp-1;p[ts]=p[tp];t[ts][a[tp]-'a']=tv;
        // add leaf ts+1. It corresponds to transition through
        c.
        t[ts][c]=ts+1;l[ts+1]=la;p[ts+1]=ts;
        // update info for the current node - remember to mark
        ts as parent of tv
        l[tp]=tp;p[tp]=ts;t[p[ts]][a[l[ts]]-'a']=ts;ts+=2;
        // prepare for descent
        // tp will mark where are we in the current suffix
        tv=s[p[ts-2]];tp=l[ts-2];
        // while the current suffix is not over, descend
        while (tp<=r[ts-2]) {tv=t[tp][a[tp]-'a'];tp+=r[tp]-l[tp
            ]+1;}
        // if we're in a node, add a suffix link to it,
        otherwise add the link to ts
        // (we'll create ts on next iteration).
        if (tp==r[ts-2]+1) s[ts-2]=tv; else s[ts-2]=ts;
        // add tp to the new edge and return to add letter to
        suffix
        tp=r[tp]-(tp-r[ts-2])+2;goto suff;
    }
}

void build() {
    ts=2;
    tv=0;
    tp=0;
    fill(r,r+N,(int)a.size()-1);
    // initialize data for the root of the tree
    s[0]=1;
```

```

l[0]=-1;
r[0]=-1;
l[1]=-1;
r[1]=-1;
memset (t, -1, sizeof t);
fill(t[1],t[1]+26,0);
// add the text to the tree, letter by letter
for (la=0; la<(int)a.size(); ++la)
    ukkadd (a[la]-'a');
}

```

SuffixArray.hpp

Description: Suffix Array.

b62f8f, 31 lines

```

struct SuffixArray{
    int n;
    vector<int> sa,isa,lcp;
    SuffixArray(){}
    SuffixArray(const string &s){init(s);}
    void init(const string &s){
        n=(int)s.size();
        sa=isa=lcp=vector<int>(n+1);
        sa[0]=n;
        iota(sa.begin()+1,sa.end(),0);
        sort(sa.begin()+1,sa.end(), [&](int i,int j){return s[i]
            <s[j];});
        for(int i=1;i<n;i++){
            int x=sa[i-1],y=sa[i];
            isa[y]=i>1&&s[x]==s[y]?isa[x]:i;
        }
        for(int len=1;len<n;len<=1){
            vector<int> ps(sa),pi(isa),pos(n+1);
            iota(pos.begin(),pos.end(),0);
            for(auto i:ps)if((i==len)>=0)sa[pos[isa[i]]++]=i;
            for(int i=1;i<n;i++){
                int x=sa[i-1],y=sa[i];
                isa[y]=pi[x]==pi[y]&&pi[x+len]==pi[y+len]?isa[x]
                    :i;
            }
        }
        for(int i=0,k=0;i<n;i++){
            for(int j=sa[isa[i]-1];j+k<n&&s[j+k]==s[i+k];k++);
            lcp[isa[i]]=k;
            if(k)k--;
        }
    }
};

```

PrefixFunction.hpp

Description: Prefix function. $\text{pi}[i]$:= the length of the longest proper prefix of $s[0:i]$ which is also a suffix of $s[0:i]$.

3d65fe, 11 lines

```

template<class STR>
vector<int> prefix_function(const STR &s){
    int n=(int)s.size();
    vector<int> pi(n);
    for(int i=1,j=0;i<n;i++){
        while(j>0&&s[i]!=s[j])j=pi[j-1];
        if(s[i]==s[j])j++;
        pi[i]=j;
    }
    return pi;
}

```

SuffixAutomaton.hpp

Description: Suffix Automaton.

Find whether a string t is a substring of a string s by traversing the automaton.

Find whether a string t is a suffix of a string s by checking whether the last node is a terminal node.

Find the number of distinct substrings of a string s by calculating the number of distinct path using DP.

Count the number of occurrences of string t in string s . Let p be the node we end up at after traversing t in the automaton. The answer is the number of paths from p to terminal nodes.

Find first occurrence of string t in string s by calculating the longest path in the automaton after reaching node p .

a50940, 49 lines

```

template<class STR>
struct SuffixAutomaton{
    using T = typename STR::value_type;
    struct Node{
        map<T,int> nxt;
        int link,len;
        Node(int link,int len):link(link),len(len){}
    };
    vector<Node> nodes;
    int last;
    SuffixAutomaton():nodes{Node(-1,0)},last(0){}
    SuffixAutomaton(const STR &s):SuffixAutomaton(){
        for(auto c:s)extend(c);
    }
    int new_node(int link,int len){
        nodes.emplace_back(Node(link,len));
        return (int)nodes.size()-1;
    }
    void extend(T c){
        int cur=new_node(0,nodes[last].len+1);
        int p=last;
        while(p!=-1&&!nodes[p].nxt.count(c)){
            nodes[p].nxt[c]=cur;
            p=nodes[p].link;
        }
        if(p!=-1){
            int q=nodes[p].nxt[c];
            if(nodes[p].len+1==nodes[q].len){
                nodes[cur].link=q;
            }else{
                int r=new_node(nodes[q].link,nodes[p].len+1);
                nodes[r].nxt=nodes[q].nxt;
                while(p!=-1&&nodes[p].nxt[c]==q){
                    nodes[p].nxt[c]=r;
                    p=nodes[p].link;
                }
                nodes[q].link=nodes[cur].link=r;
            }
        }
        last=cur;
    }
    ll distinct_substrings(){
        ll res=0;
        for(int i=1;i<(int)nodes.size();i++){
            res+=nodes[i].len-nodes[nodes[i].link].len;
        }
        return res;
    }
};

```

Geometry (9)

9.1 Geometric primitives

Point.h

Description: Class to handle points in the plane. T can be e.g. double or long long. (Avoid int.)

47ec0a, 28 lines

```

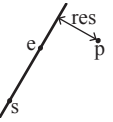
template<class T> int sgn(T x) { return (x > 0) - (x < 0); }
template<class T>
struct Point {
    typedef Point P;
    T x, y;
    explicit Point(T x=0, T y=0) : x(x), y(y) {}
    bool operator<(P p) const { return tie(x,y) < tie(p.x,p.y); }
    bool operator==(P p) const { return tie(x,y)==tie(p.x,p.y); }
    P operator+(P p) const { return P(x+p.x, y+p.y); }
    P operator-(P p) const { return P(x-p.x, y-p.y); }
    P operator*(T d) const { return P(x*d, y*d); }
    P operator/(T d) const { return P(x/d, y/d); }
    T dot(P p) const { return x*p.x + y*p.y; }
    T cross(P p) const { return x*p.y - y*p.x; }
    T cross(P a, P b) const { return (a-*this).cross(b-*this); }
    T dist2() const { return x*x + y*y; }
    double dist() const { return sqrt((double)dist2()); }
    // angle to x-axis in interval [-pi, pi]
    double angle() const { return atan2(y, x); }
    P unit() const { return *this/dist(); } // makes dist()==1
    P perp() const { return P(-y, x); } // rotates +90 degrees
    P normal() const { return perp().unit(); }
    // returns point rotated 'a' radians ccw around the origin
    P rotate(double a) const {
        return P(x*cos(a)-y*sin(a),x*sin(a)+y*cos(a)); }
    friend ostream& operator<<(ostream& os, P p) {
        return os << "(" << p.x << "," << p.y << ")"; }
};

```

lineDistance.h

Description:

Returns the signed distance between point p and the line containing points a and b . Positive value on left side and negative on right as seen from a towards b . $a==b$ gives nan. P is supposed to be `Point<T>` or `Point3D<T>` where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long. Using `Point3D` will always give a non-negative distance. For `Point3D`, call `.dist` on the result of the cross product.



"Point.h"

f6bf6b, 4 lines

```

template<class P>
double lineDist(const P& a, const P& b, const P& p) {
    return (double)(b-a).cross(p-a)/(b-a).dist();
}

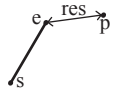
```

SegmentDistance.h

Description:

Returns the shortest distance between point p and the line segment from point s to e .

Usage: `Point<double> a, b(2,2), p(1,1);`
`bool onSegment = segDist(a,b,p) < 1e-10;`



"Point.h"

5c88f4, 6 lines

```

typedef Point<double> P;
double segDist(P& s, P& e, P& p) {
    if (s==e) return (p-s).dist();
    auto d = (e-s).dist2(), t = min(d,max(.0,(p-s).dot(e-s)));
    return ((p-s)*d-(e-s)*t).dist()/d;
}

```


SegmentIntersection.h

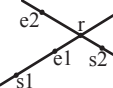
Description:
If a unique intersection point between the line segments going from s1 to e1 and from s2 to e2 exists then it is returned. If no intersection point exists an empty vector is returned. If infinitely many exist a vector with 2 elements is returned, containing the endpoints of the common line segment. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or long long.
Usage: vector<P> inter = segInter(s1,e1,s2,e2);
if (sz(inter)==1)
cout << "segments intersect at " << inter[0] << endl;
"Point.h", "OnSegment.h"9d57f2, 13 lines



```
template<class P> vector<P> segInter(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(d, b),
        oc = a.cross(b, c), od = a.cross(b, d);
    // Checks if intersection is single non-endpoint point.
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
        return {(a * ob - b * oa) / (ob - oa)};
    set<P> s;
    if (onSegment(c, d, a)) s.insert(a);
    if (onSegment(c, d, b)) s.insert(b);
    if (onSegment(a, b, c)) s.insert(c);
    if (onSegment(a, b, d)) s.insert(d);
    return {all(s)};
}
```

lineIntersection.h

Description:
If a unique intersection point of the lines going through s1,e1 and s2,e2 exists {1, point} is returned. If no intersection point exists {0, (0,0)} is returned and if infinitely many exists {-1, (0,0)} is returned. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or ll.
Usage: auto res = lineInter(s1,e1,s2,e2);
if (res.first == 1)
cout << "intersection point at " << res.second << endl;
"Point.h"ao1f81, 8 lines



```
template<class P>
pair<int, P> lineInter(P s1, P e1, P s2, P e2) {
    auto d = (e1 - s1).cross(e2 - s2);
    if (d == 0) // if parallel
        return {-(s1.cross(e1, s2) == 0), P(0, 0)};
    auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
    return {1, (s1 * p + e1 * q) / d};
}
```

sideOf.h

Description: Returns where *p* is as seen from *s* towards *e*. 1/0/-1 ⇔ left/on line/right. If the optional argument *eps* is given 0 is returned if *p* is within distance *eps* from the line. P is supposed to be Point<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long.
Usage: bool left = sideOf(p1,p2,q)==1;
"Point.h"3af81c, 9 lines

```
template<class P>
int sideOf(P s, P e, P p) { return sgn(s.cross(e, p)); }
```

```
template<class P>
int sideOf(const P& s, const P& e, const P& p, double eps) {
    auto a = (e-s).cross(p-s);
    double l = (e-s).dist()*eps;
    return (a > l) - (a < -l);
}
```

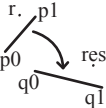
OnSegment.h

Description: Returns true iff p lies on the line segment from s to e. Use (segDist(s,e,p)<=epsilon) instead when using Point<double>.
"Point.h"cs97e8, 3 lines

```
template<class P> bool onSegment(P s, P e, P p) {
    return p.cross(s, e) == 0 && (s - p).dot(e - p) <= 0;
}
```

linearTransformation.h

Description:
Apply the linear transformation (translation, rotation and scaling) which takes line p0-p1 to line q0-q1 to point r.
"Point.h"03a306, 6 lines



```
typedef Point<double> P;
P linearTransformation(const P& p0, const P& p1,
    const P& q0, const P& q1, const P& r) {
    P dp = p1-p0, dq = q1-q0, num(dp.cross(dq), dp.dot(dq));
    return q0 + P((r-p0).cross(num), (r-p0).dot(num))/dp.dist2();
}
```

LineProjectionReflection.h

Description: Projects point p onto line ab. Set refl=true to get reflection of point p across line ab instead. The wrong point will be returned if P is an integer point and the desired point doesn't have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow.
"Point.h"b5562d, 5 lines

```
template<class P>
P lineProj(P a, P b, P p, bool refl=false) {
    P v = b - a;
    return p - v.perp()*(1+refl)*v.cross(p-a)/v.dist2();
}
```

Angle.h
Description: A class for ordering angles (as represented by int points and a number of rotations around the origin). Useful for rotational sweeping. Sometimes also represents points or vectors.
Usage: vector<Angle> v = {w[0], w[0].t360() ...}; // sorted
int j = 0; rep(i,0,n) { while (v[j] < v[i].t180()) ++j; }
// sweeps j such that (j-i) represents the number of positively oriented triangles with vertices at 0 and i
0f0602, 35 lines

```
struct Angle {
    int x, y;
    int t;
    Angle(int x, int y, int t=0) : x(x), y(y), t(t) {}
    Angle operator-(Angle b) const { return {x-b.x, y-b.y, t}; }
    int half() const {
        assert(x || y);
        return y < 0 || (y == 0 && x < 0);
    }
    Angle t90() const { return {-y, x, t + (half() && x >= 0)}; }
    Angle t180() const { return {-x, -y, t + half()}; }
    Angle t360() const { return {x, y, t + 1}; }
};

bool operator<(Angle a, Angle b) {
    // add a.dist2() and b.dist2() to also compare distances
    return make_tuple(a.t, a.half(), a.y * (ll)b.x) <
        make_tuple(b.t, b.half(), a.x * (ll)b.y);
}
```

```
// Given two points, this calculates the smallest angle between
// them, i.e., the angle that covers the defined line segment.
pair<Angle, Angle> segmentAngles(Angle a, Angle b) {
    if (b < a) swap(a, b);
    return (b < a.t180() ?
        make_pair(a, b) : make_pair(b, a.t360()));
}
```

```
}
Angle operator+(Angle a, Angle b) { // point a + vector b
    Angle r(a.x + b.x, a.y + b.y, a.t);
    if (a.t180() < r) r.t--;
    return r.t180() < a ? r.t360() : r;
}

Angle angleDiff(Angle a, Angle b) { // angle b - angle a
    int tu = b.t - a.t; a.t = b.t;
    return {a.x*b.x + a.y*b.y, a.x*b.y - a.y*b.x, tu - (b < a)};
}
```

9.2 Circles

CircleIntersection.h

Description: Computes the pair of points at which two circles intersect. Returns false in case of no intersection.
"Point.h"84d6d3, 11 lines

```
typedef Point<double> P;
bool circleInter(P a,P b,double r1,double r2,pair<P, P>* out){
    if (a == b) { assert(r1 != r2); return false; }
    P vec = b - a;
    double d2 = vec.dist2(), sum = r1+r2, dif = r1-r2,
        p = (d2 + r1*r1 - r2*r2)/(d2*2), h2 = r1*r1 - p*p*d2;
    if (sum*sum < d2 || dif*dif > d2) return false;
    P mid = a + vec*p, per = vec.perp() * sqrt(fmax(0, h2) / d2);
    *out = {mid + per, mid - per};
    return true;
}
```

CircleTangents.h

Description: Finds the external tangents of two circles, or internal if r2 is negated. Can return 0, 1, or 2 tangents - 0 if one circle contains the other (or overlaps it, in the internal case, or if the circles are the same); 1 if the circles are tangent to each other (in which case .first = .second and the tangent line is perpendicular to the line between the centers). .first and .second give the tangency points at circle 1 and 2 respectively. To find the tangents of a circle with a point set r2 to 0.
"Point.h"b0153d, 13 lines

```
template<class P>
vector<pair<P, P>> tangents(P c1, double r1, P c2, double r2) {
    P d = c2 - c1;
    double dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr;
    if (d2 == 0 || h2 < 0) return {};
    vector<pair<P, P>> out;
    for (double sign : {-1, 1}) {
        P v = (d * dr + d.perp() * sqrt(h2) * sign) / d2;
        out.push_back({c1 + v * r1, c2 + v * r2});
    }
    if (h2 == 0) out.pop_back();
    return out;
}
```

CircleLine.h

Description: Finds the intersection between a circle and a line. Returns a vector of either 0, 1, or 2 intersection points. P is intended to be Point<double>.
"Point.h"e0cfba, 9 lines

```
template<class P>
vector<P> circleLine(P c, double r, P a, P b) {
    P ab = b - a, p = a + ab * (c-a).dot(ab) / ab.dist2();
    double s = a.cross(b, c), h2 = r*r - s*s / ab.dist2();
    if (h2 < 0) return {};
    if (h2 == 0) return {p};
    P h = ab.unit() * sqrt(h2);
    return {p - h, p + h};
}
```

CirclePolygonIntersection.h

Description: Returns the area of the intersection of a circle with a ccw polygon.
Time: $\mathcal{O}(n)$

```

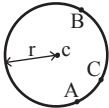
"../../../../content/geometry/Point.h"
a1ee63, 19 lines

typedef Point<double> P;
#define arg(p, q) atan2(p.cross(q), p.dot(q))
double circlePoly(P c, double r, vector<P> ps) {
    auto tri = [&](P p, P q) {
        auto r2 = r * r / 2;
        P d = q - p;
        auto a = d.dot(p)/d.dist2(), b = (p.dist2()-r*r)/d.dist2();
        auto det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        auto s = max(0., -a-sqrt(det)), t = min(1., -a+sqrt(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = p + d * t;
        return arg(p,u) * r2 + u.cross(v)/2 + arg(v,q) * r2;
    };
    auto sum = 0.0;
    rep(i,0,sz(ps))
        sum += tri(ps[i] - c, ps[(i + 1) % sz(ps)] - c);
    return sum;
}
```

circumcircle.h

Description:

The circumcircle of a triangle is the circle intersecting all three vertices. ccRadius returns the radius of the circle going through points A, B and C and ccCenter returns the center of the same circle.



```

"Point.h"
1caa3a, 9 lines

typedef Point<double> P;
double ccRadius(const P& A, const P& B, const P& C) {
    return (B-A).dist()*(C-B).dist()*(A-C).dist()/
        abs((B-A).cross(C-A))/2;
}
P ccCenter(const P& A, const P& B, const P& C) {
    P b = C-A, c = B-A;
    return A + (b*c.dist2()-c*b.dist2()).perp()/b.cross(c)/2;
}
```

MinimumEnclosingCircle.h

Description: Computes the minimum circle that encloses a set of points.
Time: expected $\mathcal{O}(n)$

```

"circumcircle.h"
09dd0a, 17 lines

pair<P, double> mec(vector<P> ps) {
    shuffle(all(ps), mt19937(time(0)));
    P o = ps[0];
    double r = 0, EPS = 1 + 1e-8;
    rep(i,0,sz(ps)) if ((o - ps[i]).dist() > r * EPS) {
        o = ps[i], r = 0;
        rep(j,0,i) if ((o - ps[j]).dist() > r * EPS) {
            o = (ps[i] + ps[j]) / 2;
            r = (o - ps[i]).dist();
            rep(k,0,j) if ((o - ps[k]).dist() > r * EPS) {
                o = ccCenter(ps[i], ps[j], ps[k]);
                r = (o - ps[i]).dist();
            }
        }
    }
    return {o, r};
}
```

9.3 Polygons

InsidePolygon.h

Description: Returns true if p lies within the polygon. If strict is true, it returns false for points on the boundary. The algorithm uses products in intermediate steps so watch out for overflow.
Usage: vector<P> v = {P{4,4}, P{1,2}, P{2,1}};
bool in = inPolygon(v, P{3, 3}, false);
Time: $\mathcal{O}(n)$

```

"Point.h", "OnSegment.h", "SegmentDistance.h"
2bf504, 11 lines

template<class P>
bool inPolygon(vector<P> &p, P a, bool strict = true) {
    int cnt = 0, n = sz(p);
    rep(i,0,n) {
        P q = p[(i + 1) % n];
        if (onSegment(p[i], q, a)) return !strict;
        //or: if (segDist(p[i], q, a) <= eps) return !strict;
        cnt ^= ((a.y<p[i].y) - (a.y<q.y)) * a.cross(p[i], q) > 0;
    }
    return cnt;
}
```

PolygonArea.h

Description: Returns twice the signed area of a polygon. Clockwise enumeration gives negative area. Watch out for overflow if using int as T!

```

"Point.h"
f12300, 6 lines

template<class T>
T polygonArea2(vector<Point<T>>& v) {
    T a = v.back().cross(v[0]);
    rep(i,0,sz(v)-1) a += v[i].cross(v[i+1]);
    return a;
}
```

PolygonCenter.h

Description: Returns the center of mass for a polygon.
Time: $\mathcal{O}(n)$

```

"Point.h"
9706dc, 9 lines

typedef Point<double> P;
P polygonCenter(const vector<P>& v) {
    P res(0, 0); double A = 0;
    for (int i = 0, j = sz(v) - 1; i < sz(v); j = i++) {
        res = res + (v[i] + v[j]) * v[j].cross(v[i]);
        A += v[j].cross(v[i]);
    }
    return res / A / 3;
}
```

PolygonCut.h

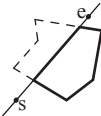
Description:

Returns a vector with the vertices of a polygon with everything to the left of the line going from s to e cut away.

```

"Point.h", "lineIntersection.h"
f2b7d4, 13 lines

typedef Point<double> P;
vector<P> polygonCut(const vector<P>& poly, P s, P e) {
    vector<P> res;
    rep(i,0,sz(poly)) {
        P cur = poly[i], prev = i ? poly[i-1] : poly.back();
        bool side = s.cross(e, cur) < 0;
        if (side != (s.cross(e, prev) < 0))
            res.push_back(lineInter(s, e, cur, prev).second);
        if (side)
            res.push_back(cur);
    }
    return res;
}
```



ConvexHull.h

Description: Returns a vector of the points of the convex hull in counter-clockwise order. Points on the edge of the hull between two other points are not considered part of the hull.
Time: $\mathcal{O}(n \log n)$



```

"Point.h"
310954, 13 lines

typedef Point<ll> P;
vector<P> convexHull(vector<P> pts) {
    if (sz(pts) <= 1) return pts;
    sort(all(pts));
    vector<P> h(sz(pts)+1);
    int s = 0, t = 0;
    for (int it = 2; it--; s = --t, reverse(all(pts)))
        for (P p : pts) {
            while (t >= s + 2 && h[t-2].cross(h[t-1], p) <= 0) t--;
            h[t++] = p;
        }
    return {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};
}
```

HullDiameter.h

Description: Returns the two points with max distance on a convex hull (ccw, no duplicate/collinear points).
Time: $\mathcal{O}(n)$

```

"Point.h"
c571b8, 12 lines

typedef Point<ll> P;
array<P, 2> hullDiameter(vector<P> S) {
    int n = sz(S), j = n < 2 ? 0 : 1;
    pair<ll, array<P, 2>> res({0, {S[0], S[0]}});
    rep(i,0,j)
        for (;;) j = (j + 1) % n) {
            res = max(res, {(S[i] - S[j]).dist2(), {S[i], S[j]}});
            if ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i]) >= 0)
                break;
        }
    return res.second;
}
```

PointInsideHull.h

Description: Determine whether a point t lies inside a convex hull (CCW order, with no collinear points). Returns true if point lies within the hull. If strict is true, points on the boundary aren't included.

```

"Point.h", "sideOf.h", "OnSegment.h"
71446b, 14 lines

typedef Point<ll> P;

bool inHull(const vector<P>& l, P p, bool strict = true) {
    int a = 1, b = sz(l) - 1, r = !strict;
    if (sz(l) < 3) return r && onSegment(l[0], l.back(), p);
    if (sideOf(l[0], l[a], l[b]) > 0) swap(a, b);
    if (sideOf(l[0], l[a], p) >= r || sideOf(l[0], l[b], p) <= -r)
        return false;
    while (abs(a - b) > 1) {
        int c = (a + b) / 2;
        (sideOf(l[0], l[c], p) > 0 ? b : a) = c;
    }
    return sgn(l[a].cross(l[b], p)) < r;
}
```

LineHullIntersection.h

Description: Line-convex polygon intersection. The polygon must be ccw and have no collinear points. lineHull(line, poly) returns a pair describing the intersection of a line with the polygon: $\bullet(-1, -1)$ if no collision, $\bullet(i, -1)$ if touching the corner i , $\bullet(i, i)$ if along side $(i, i+1)$, $\bullet(i, j)$ if crossing sides $(i, i+1)$ and $(j, j+1)$. In the last case, if a corner i is crossed, this is treated as happening on side $(i, i+1)$. The points are returned in the same order as the line hits the polygon. extrVertex returns the point of a hull with the max projection onto a line.
Time: $\mathcal{O}(\log n)$

```
"Point.h" 7cf45b, 39 lines

#define cmp(i,j) sgn(dir.perp().cross(poly[(i)%n]-poly[(j)%n]))
#define extr(i) cmp(i+1, i) >= 0 && cmp(i, i-1+n) < 0
template <class P> int extrVertex(vector<P>& poly, P dir) {
    int n = sz(poly), lo = 0, hi = n;
    if (extr(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (extr(m)) return m;
        int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi : lo) = m;
    }
    return lo;
}

#define cmpL(i) sgn(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
    int endA = extrVertex(poly, (a - b).perp());
    int endB = extrVertex(poly, (b - a).perp());
    if (cmpL(endA) < 0 || cmpL(endB) > 0)
        return {-1, -1};
    array<int, 2> res;
    rep(i,0,2) {
        int lo = endB, hi = endA, n = sz(poly);
        while ((lo + 1) % n != hi) {
            int m = ((lo + hi + (lo < hi ? 0 : n)) / 2) % n;
            (cmpL(m) == cmpL(endB) ? lo : hi) = m;
        }
        res[i] = (lo + !cmpL(hi)) % n;
        swap(endA, endB);
    }
    if (res[0] == res[1]) return {res[0], -1};
    if (!cmpL(res[0]) && !cmpL(res[1]))
        switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly)) {
            case 0: return {res[0], res[0]};
            case 2: return {res[1], res[1]};
        }
    return res;
}
```

9.4 Misc. Point Set Problems

ClosestPair.h

Description: Finds the closest pair of points.
Time: $\mathcal{O}(n \log n)$

```
"Point.h" ac41a6, 17 lines

typedef Point<ll> P;
pair<P, P> closest(vector<P> v) {
    assert(sz(v) > 1);
    set<P> S;
    sort(all(v), [](P a, P b) { return a.y < b.y; });
    pair<ll, pair<P, P>> ret{LLONG_MAX, {P(), P()}};
    int j = 0;
    for (P p : v) {
        P d{1 + (ll)sqrt(ret.first), 0};
        while (v[j].y <= p.y - d.x) S.erase(v[j++]);
        auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
        for (; lo != hi; ++lo)
            ret = min(ret, {(lo - p).dist2(), {lo, p}});
    }
```

```
S.insert(p);
}
return ret.second;
}
```

ManhattanMST.h

Description: Given N points, returns up to $4 \cdot N$ edges, which are guaranteed to contain a minimum spanning tree for the graph with edge weights $w(p, q) = |p.x - q.x| + |p.y - q.y|$. Edges are in the form (distance, src, dst). Use a standard MST algorithm on the result to find the final MST.
Time: $\mathcal{O}(N \log N)$

```
"Point.h" df6f59, 23 lines

typedef Point<int> P;
vector<array<int, 3>> manhattanMST(vector<P> ps) {
    vi id(sz(ps));
    iota(all(id), 0);
    vector<array<int, 3>> edges;
    rep(k,0,4) {
        sort(all(id), [&](int i, int j) {
            return (ps[i]-ps[j]).x < (ps[j]-ps[i]).y;});
        map<int, int> sweep;
        for (int i : id) {
            for (auto it = sweep.lower_bound(-ps[i].y);
                it != sweep.end(); sweep.erase(it++)) {
                int j = it->second;
                P d = ps[i] - ps[j];
                if (d.y > d.x) break;
                edges.push_back({d.y + d.x, i, j});
            }
            sweep[-ps[i].y] = i;
        }
        for (P& p : ps) if (k & 1) p.x = -p.x; else swap(p.x, p.y);
    }
    return edges;
}
```

kdTree.h

Description: KD-tree (2d, can be extended to 3d)

```
"Point.h" bac5b0, 63 lines

typedef long long T;
typedef Point<T> P;
const T INF = numeric_limits<T>::max();

bool on_x(const P& a, const P& b) { return a.x < b.x; }
bool on_y(const P& a, const P& b) { return a.y < b.y; }

struct Node {
    P pt; // if this is a leaf, the single point in it
    T x0 = INF, x1 = -INF, y0 = INF, y1 = -INF; // bounds
    Node *first = 0, *second = 0;

    T distance(const P& p) { // min squared distance to a point
        T x = (p.x < x0 ? x0 : p.x > x1 ? x1 : p.x);
        T y = (p.y < y0 ? y0 : p.y > y1 ? y1 : p.y);
        return (P(x,y) - p).dist2();
    }

    Node(vector<P>&& vp) : pt(vp[0]) {
        for (P p : vp) {
            x0 = min(x0, p.x); x1 = max(x1, p.x);
            y0 = min(y0, p.y); y1 = max(y1, p.y);
        }
        if (vp.size() > 1) {
            // split on x if width >= height (not ideal...)
            sort(all(vp), x1 - x0 >= y1 - y0 ? on_x : on_y);
            // divide by taking half the array for each child (not
            // best performance with many duplicates in the middle)
            int half = sz(vp)/2;
```

```
        first = new Node({vp.begin(), vp.begin() + half});
        second = new Node({vp.begin() + half, vp.end()});
    }
};

struct KDTree {
    Node* root;
    KDTree(const vector<P>& vp) : root(new Node({all(vp)})) {}

    pair<T, P> search(Node *node, const P& p) {
        if (!node->first) {
            // uncomment if we should not find the point itself:
            // if (p == node->pt) return {INF, P()};
            return make_pair((p - node->pt).dist2(), node->pt);
        }

        Node *f = node->first, *s = node->second;
        T bfirst = f->distance(p), bsec = s->distance(p);
        if (bfirst > bsec) swap(bsec, bfirst), swap(f, s);

        // search closest side first, other side if needed
        auto best = search(f, p);
        if (bsec < best.first)
            best = min(best, search(s, p));
        return best;
    }

    // find nearest point to a point, and its squared distance
    // (requires an arbitrary operator< for Point)
    pair<T, P> nearest(const P& p) {
        return search(root, p);
    }
};
```

DelaunayTriangulation.h

Description: Computes the Delaunay triangulation of a set of points. Each circumcircle contains none of the input points. If any three points are collinear or any four are on the same circle, behavior is undefined.
Time: $\mathcal{O}(n^2)$

```
"Point.h", "3dHull.h" c0e7bc, 10 lines

template<class P, class F>
void delaunay(vector<P>& ps, F trifun) {
    if (sz(ps) == 3) { int d = (ps[0].cross(ps[1], ps[2]) < 0);
        trifun(0,1+d,2-d); }
    vector<P3> p3;
    for (P p : ps) p3.emplace_back(p.x, p.y, p.dist2());
    if (sz(ps) > 3) for(auto t:hull3d(p3)) if ((p3[t.b]-p3[t.a]).
        cross(p3[t.c]-p3[t.a]).dot(P3(0,0,1)) < 0)
        trifun(t.a, t.c, t.b);
}
```

FastDelaunay.h

Description: Fast Delaunay triangulation. Each circumcircle contains none of the input points. There must be no duplicate points. If all points are on a line, no triangles will be returned. Should work for doubles as well, though there may be precision issues in 'circ'. Returns triangles in order {t[0][0], t[0][1], t[0][2], t[1][0], ...}, all counter-clockwise.
Time: $\mathcal{O}(n \log n)$

```
"Point.h" eefdf5, 88 lines

typedef Point<ll> P;
typedef struct Quad* Q;
typedef __int128_t lll; // (can be ll if coords are < 2e4)
P arb(LLONG_MAX,LLONG_MAX); // not equal to any other point

struct Quad {
    Q rot, o; P p = arb; bool mark;
    P& F() { return r()->p; }
```

```
Q& r() { return rot->rot; }
Q prev() { return rot->o->rot; }
Q next() { return r()->prev(); }
} *H;

bool circ(P p, P a, P b, P c) { // is p in the circumcircle?
    ll p2 = p.dist2(), A = a.dist2()-p2,
        B = b.dist2()-p2, C = c.dist2()-p2;
    return p.cross(a,b)*C + p.cross(b,c)*A + p.cross(c,a)*B > 0;
}

Q makeEdge(P orig, P dest) {
    Q r = H ? H : new Quad{new Quad{new Quad{0}}};
    H = r->o; r->r()->r() = r;
    rep(i,0,4) r = r->rot, r->p = arb, r->o = i & 1 ? r : r->r();
    r->p = orig; r->F() = dest;
    return r;
}

void splice(Q a, Q b) {
    swap(a->o->rot->o, b->o->rot->o); swap(a->o, b->o);
}

Q connect(Q a, Q b) {
    Q q = makeEdge(a->F(), b->p);
    splice(q, a->next());
    splice(q->r(), b);
    return q;
}

pair<Q,Q> rec(const vector<P>& s) {
    if (sz(s) <= 3) {
        Q a = makeEdge(s[0], s[1]), b = makeEdge(s[1], s.back());
        if (sz(s) == 2) return { a, a->r() };
        splice(a->r(), b);
        auto side = s[0].cross(s[1], s[2]);
        Q c = side ? connect(b, a) : 0;
        return {side < 0 ? c->r() : a, side < 0 ? c : b->r() };
    }

#define H(e) e->F(), e->p
#define valid(e) (e->F().cross(H(base)) > 0)
    Q A, B, ra, rb;
    int half = sz(s) / 2;
    tie(ra, A) = rec({all(s) - half});
    tie(B, rb) = rec({sz(s) - half + all(s)});
    while ((B->p.cross(H(A)) < 0 && (A = A->next())) ||
        (A->p.cross(H(B)) > 0 && (B = B->r()->o)));
    Q base = connect(B->r(), A);
    if (A->p == ra->p) ra = base->r();
    if (B->p == rb->p) rb = base;

#define DEL(e, init, dir) Q e = init->dir; if (valid(e)) \
    while (circ(e->dir->F(), H(base), e->F())) { \
        Q t = e->dir; \
        splice(e, e->prev()); \
        splice(e->r(), e->r()->prev()); \
        e->o = H; H = e; e = t; \
    }
    for (;;) {
        DEL(LC, base->r(), o); DEL(RC, base, prev());
        if (!valid(LC) && !valid(RC)) break;
        if (!valid(LC) || (valid(RC) && circ(H(RC), H(LC))))
            base = connect(RC, base->r());
        else
            base = connect(base->r(), LC->r());
    }
    return { ra, rb };
}

vector<P> triangulate(vector<P> pts) {
    sort(all(pts)); assert(unique(all(pts)) == pts.end());
```

```
if (sz(pts) < 2) return {};
Q e = rec(pts).first;
vector<Q> q = {e};
int qi = 0;
while (e->o->F().cross(e->F(), e->p) < 0) e = e->o;
#define ADD { Q c = e; do { c->mark = 1; pts.push_back(c->p); \
    q.push_back(c->r()); c = c->next(); } while (c != e); }
    ADD; pts.clear();
    while (qi < sz(q)) if (!(e = q[qi++])->mark) ADD;
    return pts;
}
```

9.5 3D

PolyhedronVolume.h

Description: Magic formula for the volume of a polyhedron. Faces should point outwards.

```
template<class V, class L>
double signedPolyVolume(const V& p, const L& trilst) {
    double v = 0;
    for (auto i : trilst) v += p[i.a].cross(p[i.b]).dot(p[i.c]);
    return v / 6;
}
```

Point3D.h

Description: Class to handle points in 3D space. T can be e.g. double or long long.

```
template<class T> struct Point3D {
    typedef Point3D P;
    typedef const P& R;
    T x, y, z;
    explicit Point3D(T x=0, T y=0, T z=0) : x(x), y(y), z(z) {}
    bool operator<(R p) const {
        return tie(x, y, z) < tie(p.x, p.y, p.z); }
    bool operator==(R p) const {
        return tie(x, y, z) == tie(p.x, p.y, p.z); }
    P operator+(R p) const { return P(x+p.x, y+p.y, z+p.z); }
    P operator-(R p) const { return P(x-p.x, y-p.y, z-p.z); }
    P operator*(T d) const { return P(x*d, y*d, z*d); }
    P operator/(T d) const { return P(x/d, y/d, z/d); }
    T dot(R p) const { return x*p.x + y*p.y + z*p.z; }
    P cross(R p) const {
        return P(y*p.z - z*p.y, z*p.x - x*p.z, x*p.y - y*p.x);
    }
    T dist2() const { return x*x + y*y + z*z; }
    double dist() const { return sqrt((double)dist2()); }
    //Azimuthal angle (longitude) to x-axis in interval [-pi, pi]
    double phi() const { return atan2(y, x); }
    //Zenith angle (latitude) to the z-axis in interval [0, pi]
    double theta() const { return atan2(sqrt(x*x+y*y),z); }
    P unit() const { return *this/(T)dist(); } //makes dist()==1
    //returns unit vector normal to *this and p
    P normal(P p) const { return cross(p).unit(); }
    //returns point rotated 'angle' radians ccw around axis
    P rotate(double angle, P axis) const {
        double s = sin(angle), c = cos(angle); P u = axis.unit();
        return u.dot(u)*(1-c) + (*this)*c - cross(u)*s;
    }
};
```

3dHull.h

Description: Computes all faces of the 3-dimension hull of a point set. *No four points must be coplanar*, or else random results will be returned. All faces will point outwards.

Time: $\mathcal{O}(n^2)$

```
"Point3D.h"
typedef Point3D<double> P3;
```

```
struct PR {
    void ins(int x) { (a == -1 ? a : b) = x; }
    void rem(int x) { (a == x ? a : b) = -1; }
    int cnt() { return (a != -1) + (b != -1); }
    int a, b;
};
```

```
struct F { P3 q; int a, b, c; };
```

```
vector<F> hull3d(const vector<P3>& A) {
    assert(sz(A) >= 4);
    vector<vector<PR>> E(sz(A), vector<PR>(sz(A), {-1, -1}));
#define E(x,y) E[f.x][f.y]
    vector<F> FS;
    auto mf = [&](int i, int j, int k, int l) {
        P3 q = (A[j] - A[i]).cross((A[k] - A[i]));
        if (q.dot(A[l]) > q.dot(A[i]))
            q = q * -1;
        F f{q, i, j, k};
        E(a,b).ins(k); E(a,c).ins(j); E(b,c).ins(i);
        FS.push_back(f);
    };
    rep(i,0,4) rep(j,i+1,4) rep(k,j+1,4)
        mf(i, j, k, 6 - i - j - k);
```

```
rep(i,4,sz(A)) {
    rep(j,0,sz(FS)) {
        F f = FS[j];
        if (f.q.dot(A[i]) > f.q.dot(A[f.a])) {
            E(a,b).rem(f.c);
            E(a,c).rem(f.b);
            E(b,c).rem(f.a);
            swap(FS[j--], FS.back());
            FS.pop_back();
        }
    }
    int nw = sz(FS);
    rep(j,0,nw) {
        F f = FS[j];
#define C(a, b, c) if (E(a,b).cnt() != 2) mf(f.a, f.b, i, f.c);
        C(a, b, c); C(a, c, b); C(b, c, a);
    }
    for (F& it : FS) if ((A[it.b] - A[it.a]).cross(
        A[it.c] - A[it.a]).dot(it.q) <= 0) swap(it.c, it.b);
    return FS;
};
```

sphericalDistance.h

Description: Returns the shortest distance on the sphere with radius radius between the points with azimuthal angles (longitude) f1 (ϕ_1) and f2 (ϕ_2) from x axis and zenith angles (latitude) t1 (θ_1) and t2 (θ_2) from z axis (0 = north pole). All angles measured in radians. The algorithm starts by converting the spherical coordinates to cartesian coordinates so if that is what you have you can use only the two last rows. dx*radius is then the difference between the two points in the x direction and d*radius is the total distance between the points.

```
double sphericalDistance(double f1, double t1,
    double f2, double t2, double radius) {
    double dx = sin(t2)*cos(f2) - sin(t1)*cos(f1);
    double dy = sin(t2)*sin(f2) - sin(t1)*sin(f1);
    double dz = cos(t2) - cos(t1);
    double d = sqrt(dx*dx + dy*dy + dz*dz);
    return radius*2*asin(d/2);
}
```

ComplexGeometry.hpp

Description: geometry using std complex

9a240d, 33 lines

```
using namespace std;

using point = complex<double>;

double dot(const point &a,const point &b){ return real(conj(a)*
b); }
double cross(const point &a,const point &b){ return imag(conj(a)
*b); }

point rotate_by(const point &p,const point about, double
radians){
    return (p-about)*exp(point(0,radians))+about;
}

point project(const point &p,const point &about1,const point &
about2){
    point z=p-about1;
    point w=about2-about1;
    return w*dot(z,w)/norm(w)+about1;
}

point reflex(const point &p,const point &about1,const point &
about2){
    point z=p-about1;
    point w=about2-about1;
    return conj(z/w)*w+about1;
}

point intersect(const point &a,const point &b,const point &p,
const point &q){
    double d1=cross(p-a,b-a);
    double d2=cross(q-a,b-a);
    return (d1*q-d2*p)/(d1-d2); // undefined if they are
parallel

// find angle abc
point angle(const point &a,const point &b,const point &c){
    return abs(remainder(arg(a-b)-arg(c-b),2.0*M_PI));
}
```

DefiniteIntegral.hpp

Description: Definite integral using Simpson's formula.

edbaf5, 15 lines

```
template<class T,class F>
T definite_integral(T a,T b,const F &f,int n){
    T res=0;
    T dx=(b-a)/n;
    T fl=0,fr=f(a);
    for(int i=0;i<n;i++){
        T l=a+dx*i;
        T r=l+dx;
        fl=fr;
        fr=f(r);
        T fm=f((l+r)/2);
        res+=fl+4*fm+fr;
    }
    return res*dx/6;
}
```

CHT.hpp

Description: Convex Hull Container

5012e6, 23 lines

```
struct CHT {
    struct Line {
        ll m, c;
        Line (ll _m, ll _c) { m = _m, c = _c; }
        ll eval(ll x) { return m*x + c; }
    };
    deque<Line> dq;

    bool check(Line base, Line cur, Line add) {
        return (cur.c - base.c)*(base.m - add.m) < (add.c - base.c)
            *(base.m - cur.m);
    }

    void insert(ll m, ll c) {
        Line l(m, c);
        while (dq.size() > 1 && !check(dq.end() [-2], dq.back(), l))
            dq.pop_back();
        dq.push_back(l);
    }

    ll query(ll x) {
        while (dq.size() > 1 && dq[0].eval(x) < dq[1].eval(x)) dq.
            pop_front();
        return dq[0].eval(x);
    }
};
```

Knuth.hpp

Description: Knuth

04ace3, 30 lines

```
int solve() {
    int N;
    ... // read N and input
    int dp[N][N], opt[N][N];

    auto C = [&](int i, int j) {
        ... // Implement cost function C.
    };

    for (int i = 0; i < N; i++) {
        opt[i][i] = i;
        ... // Initialize dp[i][i] according to the problem
    }

    for (int i = N-2; i >= 0; i--) {
        for (int j = i+1; j < N; j++) {
            int mn = INT_MAX;
            int cost = C(i, j);
            for (int k = opt[i][j-1]; k <= min(j-1, opt[i+1][j]
                )); k++) {
                if (mn >= dp[i][k] + dp[k+1][j] + cost) {
                    opt[i][j] = k;
                    mn = dp[i][k] + dp[k+1][j] + cost;
                }
            }
            dp[i][j] = mn;
        }
    }

    return dp[0][N-1];
}
```

DVC.hpp

Description: Optimize $O(N^2K)$ to $O(NK \log N)$

aa5ddf, 19 lines

```
vector<vl> cst, dp;
```

```
ll cost(int l, int r) {
    return cst[l][r];
}

void divide(int l, int r, int opt_l, int opt_r, int c) {
    if(l > r) return ;
    int mid = (l + r) / 2;
    pair<ll, int> best = make_pair(INF, -1);
    for(int k=opt_l; k<=min(mid, opt_r); ++k) {
        best = min(best, make_pair(dp[c - 1][k] + cost(k + 1, mid),
            k));
    }
    dp[c][mid] = best.first;
    divide(l, mid - 1, opt_l, best.second, c);
    divide(mid + 1, r, best.second, opt_r, c);
}

// for(int c=1; c<=K; ++c) divide(1, N, 1, N, c);
```

SlopeTrick.hpp

Description: Absolute Smth

f62f9a, 36 lines

```
ll extending_value;

struct slope_trick {
    multiset<ll> ms_l, ms_r;
    ll min_y = 0ll, lz_l = 0ll, lz_r = 0ll;
    bool extending = false;
    void add_line(ll v) {
        if(extending) {
            lz_l -= extending_value;
            lz_r -= extending_value;
        }
        extending = true;
        if(ms_l.empty() && ms_r.empty()) {
            ms_l.emplace(v);
            ms_r.emplace(v);
        }
        else if(v <= *ms_l.rbegin() + lz_l) {
            min_y += (*ms_l.rbegin() + lz_l) - v;
            ms_r.emplace(*ms_l.rbegin() + lz_l - lz_r);
            ms_l.erase(--ms_l.end());
            ms_l.emplace(v - lz_l);
            ms_l.emplace(v - lz_l);
        }
        else if(v >= *ms_r.begin() + lz_r) {
            min_y += v - (*ms_r.begin() + lz_r);
            ms_l.emplace(*ms_r.begin() + lz_r - lz_l);
            ms_r.erase(ms_r.begin());
            ms_r.emplace(v - lz_r);
            ms_r.emplace(v - lz_r);
        }
        else {
            ms_l.emplace(v - lz_l);
            ms_r.emplace(v - lz_r);
        }
    }
};
```

AlienTrick.hpp

Description: Alien Trick

8435cb, 53 lines

```
using namespace std;

using ll=long long;
using tl=__int128_t;
using db=double;
```

```
const ll N=500005;
ll c[N],sum[N];
deque<tuple<ll,ll,int>> dq;

ll minPenguinValue(int n, int m, vector<int> a){
    ll l=0,r=2e18,re;
    for (int i=1;i<=n;i++){
        sum[i]=sum[i-1]+ll(a[i-1]);
        c[i]=c[i-1]+sum[i]*ll(a[i-1]);
    }
    while (l<=r){
        ll mid=l+(r-l)/2;
        while (!dq.empty()) dq.pop_back();
        dq.emplace_back(0,0,0);
        for (int i=1;i<=n;i++){
            while (dq.size()>1){
                auto [m1,c1,u1]=dq.front();
                dq.pop_front();
                auto [m2,c2,u2]=dq.front();
                if (c2-c1>sum[i]*(m1-m2)){
                    dq.emplace_front(m1,c1,u1);
                    break;
                }
            }
            auto [nm,nc,nu]=dq.front();
            ll nv=nm*sum[i]+nc+c[i]+mid;
            ll m3=-sum[i],c3=-c[i]+sum[i]*sum[i]+nv;
            if (i==n){
                if (nu+1>m) l=mid+1;
                else{
                    re=nv-m*mid;
                    r=mid-1;
                }
            }
            while (dq.size()>1){
                auto [m2,c2,u2]=dq.back();
                dq.pop_back();
                auto [m1,c1,u1]=dq.back();
                if (db(c3-c1)/db(m1-m3)<=db(c3-c2)/db(m2-m3)){
                    dq.emplace_back(m2,c2,u2);
                    break;
                }
            }
            dq.emplace_back(m3,c3,nu+1);
        }
    }
    return re;
}
```

Polynomials (11)

11.1 Newton’s Method

if $F(Q) = 0$, then $Q_{2n} \equiv Q_n - \frac{F(Q_n)}{F'(Q_n)} \pmod{x^{2n}}$

$$Q = P^{-1} : Q_{2n} \equiv Q_n \cdot (2 - P \cdot Q_n^2) \pmod{x^{2n}}$$

$$Q = \ln P = \int \frac{P'}{P} dx$$

$$Q = e^P : Q_{2n} \equiv Q_n(1 + P - \ln Q_n) \pmod{x^{2n}}$$

$$Q = \sqrt{P} : Q_{2n} \equiv \frac{1}{2}(Q_n + P \cdot Q_n^{-1}) \pmod{x^{2n}}$$

$$Q = P^k = \alpha^k x^{kt} e^{k \ln T}; P = \alpha \cdot x^t \cdot T, T(0) = 1$$

Interpolation FormalPowerSeries

Interpolation.hpp

Description: Given n points $(x[i], y[i])$, computes an $n-1$ -degree polynomial p that passes through them: $p(x) = a[0] * x^0 + \dots + a[n-1] * x^{n-1}$. For numerical precision, pick $x[k] = c * \cos(k/(n-1) * \pi), k = 0 \dots n-1$.
Time: $\mathcal{O}(n^2)$

```
typedef vector<double> vd;
vd interpolate(vd x, vd y, int n) {
    vd res(n), temp(n);
    rep(k,0,n-1) rep(i,k+1,n)
        y[i] = (y[i] - y[k]) / (x[i] - x[k]);
    double last = 0; temp[0] = 1;
    rep(k,0,n) rep(i,0,n) {
        res[i] += y[k] * temp[i];
        swap(last, temp[i]);
        temp[i] -= last * x[k];
    }
    return res;
}
```

FormalPowerSeries.hpp

Description: basic operations of formal power series

```
template<class mint>
struct FormalPowerSeries:vector<mint>{
    using vector<mint>::vector;
    using FPS = FormalPowerSeries;

    FPS &operator+=(const FPS &rhs){
        if(rhs.size()>this->size())this->resize(rhs.size());
        for(int i=0;i<rhs.size();i++) (*this)[i]+=rhs[i];
        return *this;
    }
    FPS &operator+=(const mint &rhs){
        if(this->empty())this->resize(1);
        (*this)[0]+=rhs;
        return *this;
    }
    FPS &operator-=(const FPS &rhs){
        if(rhs.size()>this->size())this->resize(rhs.size());
        for(int i=0;i<rhs.size();i++) (*this)[i]-=rhs[i];
        return *this;
    }
    FPS &operator-=(const mint &rhs){
        if(this->empty())this->resize(1);
        (*this)[0]-=rhs;
        return *this;
    }
    // FPS &operator*=(const FPS &rhs){
    //     auto res=NTT<mint>()( *this, rhs );
    //     return *this=FPS(res.begin(), res.end());
    // }
    FPS &operator*=(const mint &rhs){
        for(auto &a:*this)a*=rhs;
        return *this;
    }
    friend FPS operator+(FPS lhs,const FPS &rhs){return lhs+=rhs;}
    friend FPS operator+(FPS lhs,const mint &rhs){return lhs+=rhs;}
    friend FPS operator+(const mint &lhs,FPS &rhs){return rhs+=lhs;}
    friend FPS operator-(FPS lhs,const FPS &rhs){return lhs-=rhs;}
    friend FPS operator-(FPS lhs,const mint &rhs){return lhs-=rhs;}
    friend FPS operator-(const mint &lhs,FPS rhs){return -(rhs-lhs);}
```

```
friend FPS operator*(FPS lhs,const FPS &rhs){return lhs*=rhs;}
friend FPS operator*(FPS lhs,const mint &rhs){return lhs*=rhs;}
friend FPS operator*(const mint &lhs,FPS rhs){return rhs*=lhs;}

FPS operator-(){return (*this)*-1;}

FPS rev(){
    FPS res(*this);
    reverse(res.begin(),res.end());
    return res;
}
FPS pre(int sz){
    FPS res(this->begin(),this->begin()+min((int)this->size(),sz));
    if(res.size()<sz)res.resize(sz);
    return res;
}
FPS shrink(){
    FPS res(*this);
    while(!res.empty()&&res.back()==mint{})res.pop_back();
    return res;
}
FPS operator>>(int sz){
    if(this->size()<=sz)return {};
    FPS res(*this);
    res.erase(res.begin(),res.begin()+sz);
    return res;
}
FPS operator<<(int sz){
    FPS res(*this);
    res.insert(res.begin(),sz,mint{});
    return res;
}
FPS diff(){
    const int n=this->size();
    FPS res(max(0,n-1));
    for(int i=1;i<n;i++)res[i-1]=(*this)[i]*mint(i);
    return res;
}
FPS integral(){
    const int n=this->size();
    FPS res(n+1);
    res[0]=0;
    if(n>0)res[1]=1;
    ll mod=mint::get_mod();
    for(int i=2;i<=n;i++)res[i]=(-res[mod%i])*(mod/i);
    for(int i=0;i<n;i++)res[i+1]=(*this)[i];
    return res;
}
mint eval(const mint &x){
    mint res=0,w=1;
    for(auto &a:*this)res+=a*w,w*=x;
    return res;
}

FPS inv(int deg=-1){
    assert(!this->empty()&&(*this)[0]!=mint(0));
    if(deg===-1)deg=this->size();
    FPS res{mint(1)/(*this)[0]};
    for(int i=2;i>1<deg;i<=1){
        res=(res*(mint(2)-res*pre(i))).pre(i);
    }
    return res.pre(deg);
}
FPS log(int deg=-1){
    assert(!this->empty()&&(*this)[0]==mint(1));
```

```

        if (deg==-1)deg=this->size();
        return (pre(deg).diff()*inv(deg)).pre(deg-1).integral()
        ;
    }
    FPS exp(int deg=-1){
        assert(this->empty()||(*this)[0]==mint(0));
        if (deg==-1)deg=this->size();
        FPS res{mint(1)};
        for(int i=2;i>>1<deg;i<<=1){
            res=(res*(pre(i)-res.log(i)+mint(1))).pre(i);
        }
        return res.pre(deg);
    }
    FPS pow(ll k,int deg=-1){
        const int n=this->size();
        if (deg==-1)deg=n;
        if (k==0){
            FPS res(deg);
            if (deg)res[0]=mint(1);
            return res;
        }
        for(int i=0;i<n;i++){
            if(__int128_t(i)*k>=deg)return FPS(deg,mint(0));
            if((*this)[i]==mint(0))continue;
            mint rev=mint(1)/(*this)[i];
            FPS res=(((*this*rev)>>i).log(deg)*k).exp(deg);
            res=((res*binpow((*this)[i],k)<<(i*k)).pre(deg);
            return res;
        }
        return FPS(deg,mint(0));
    }
};
using FPS=FormalPowerSeries<mint>;

```

FFTNTT.hpp

Description: basic operations of formal power series

db0936, 222 lines

```

using ll = long long;
using poly = vector<ll>;
const ll M=998244353; // M=7340033 -> pr=5, M=998244353 -> pr=3
ll pr=3,inv_pr,m;

int qp(int a, int e = M - 2) {
    ll r = 1, x = a % M;
    while (e > 0) {if (e & 1) r = r * x % M;x = x * x % M;e >>= 1;}
    return (int)r;
}

void ntt(poly &a, bool inv) {
    int n = a.size();
    static poly rev, roots{0,1};
    if ((int)rev.size() != n) {
        int L = __builtin_ctz(n);
        rev.assign(n,0);
        for (int i = 0; i < n; i++) rev[i] = (rev[i>>1]>>1) | ((i&1)<<(L-1));
    }
    for (int i = 0; i < n; i++) if (i<rev[i]) swap(a[i], a[rev[i]]);

    if ((int)roots.size() < n) {
        int k = __builtin_ctz(roots.size());
        roots.resize(n);
        while ((1<<k) < n) {
            int e = qp(pr, (M-1) >> (k+1));
            for (int i = 1<<(k-1); i < (1<<k); i++) {
                roots[2*i] = roots[i];
                roots[2*i+1] = (ll)roots[i] * e % M;
            }
        }
    }
    for (int len = 1; len < n; len <<= 1) {
        for (int i = 0; i < n; i += 2*len) {
            for (int j = 0; j < len; j++) {
                int u = a[i+j];
                int v = (ll)a[i+j+len] * roots[len+j] % M;
                a[i+j] = u + v < M ? u + v : u + v - M;
                a[i+j+len] = u - v >= 0 ? u - v : u - v + M;
            }
        }
    }
    if (inv) {
        reverse(a.begin()+1, a.end());
        int inv_n = qp(n);
        for (auto &x : a) x = (ll)x * inv_n % M;
    }
}

using cd = complex<double>; //a + bi (real,imag)
const double PI = acos(-1);

void fft(vector<cd> &a, bool invert) {
    int n = a.size();

    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1)
            j ^= bit;
        j ^= bit;

        if (i < j)
            swap(a[i], a[j]);
    }

    for (int len = 2; len <= n; len <<= 1) {
        double ang = 2 * PI / len * (invert ? -1 : 1);
        cd wlen(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            cd w(1);
            for (int j = 0; j < len / 2; j++) {
                cd u = a[i+j], v = a[i+j+len/2] * w;
                a[i+j] = u + v;
                a[i+j+len/2] = u - v;
                w *= wlen;
            }
        }
    }
    if (invert) {
        for (cd &x : a)
            x /= n;
    }
}

poly multiply(poly a, poly b) {
    int sz = a.size() + b.size() - 1;
    int n = 1; while (n < sz) n <<= 1;
    a.resize(n); b.resize(n);
    ntt(a, false); ntt(b, false);
    for (int i = 0; i < n; i++) a[i] = (ll)a[i] * b[i] % M;
    ntt(a, true);
    a.resize(sz);
    return a;
}

```

```

        }
        k++;
    }
}

for (int len = 1; len < n; len <<= 1) {
    for (int i = 0; i < n; i += 2*len) {
        for (int j = 0; j < len; j++) {
            int u = a[i+j];
            int v = (ll)a[i+j+len] * roots[len+j] % M;
            a[i+j] = u + v < M ? u + v : u + v - M;
            a[i+j+len] = u - v >= 0 ? u - v : u - v + M;
        }
    }
}

if (inv) {
    reverse(a.begin()+1, a.end());
    int inv_n = qp(n);
    for (auto &x : a) x = (ll)x * inv_n % M;
}

using cd = complex<double>; //a + bi (real,imag)
const double PI = acos(-1);

void fft(vector<cd> &a, bool invert) {
    int n = a.size();

    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1)
            j ^= bit;
        j ^= bit;

        if (i < j)
            swap(a[i], a[j]);
    }

    for (int len = 2; len <= n; len <<= 1) {
        double ang = 2 * PI / len * (invert ? -1 : 1);
        cd wlen(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            cd w(1);
            for (int j = 0; j < len / 2; j++) {
                cd u = a[i+j], v = a[i+j+len/2] * w;
                a[i+j] = u + v;
                a[i+j+len/2] = u - v;
                w *= wlen;
            }
        }
    }
    if (invert) {
        for (cd &x : a)
            x /= n;
    }
}

poly multiply(poly a, poly b) {
    int sz = a.size() + b.size() - 1;
    int n = 1; while (n < sz) n <<= 1;
    a.resize(n); b.resize(n);
    ntt(a, false); ntt(b, false);
    for (int i = 0; i < n; i++) a[i] = (ll)a[i] * b[i] % M;
    ntt(a, true);
    a.resize(sz);
    return a;
}

```

```

vector<cd> multiply(vector<cd> a, vector<cd> b) {
    int sz = a.size() + b.size() - 1;
    int n = 1; while (n < sz) n <<= 1;
    a.resize(n); b.resize(n);
    fft(a, false); fft(b, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i];
    fft(a, true);
    a.resize(sz);
    return a;
}

poly poly_inv(poly f, int m) {
    poly g(1, qp(f[0])); // g0 = 1/f0
    for (int len = 1; len < m; len <<= 1) {
        poly f_cut(f.begin(), f.begin() + min((int)f.size(), 2*len));
        auto tmp = multiply(multiply(g, g), f_cut);
        tmp.resize(2*len);
        g.resize(2*len);
        for (int i = 0; i < 2*len; i++) {
            g[i] = ((2LL*g[i] - tmp[i]) % M + M) % M;
        }
    }
    g.resize(m);
    return g;
}

poly poly_sqrt(const poly& f, int m) {
    poly g(1,1);
    int inv2 = (M + 1) / 2;
    for (int len = 1; len < m; len <<= 1) {
        auto g2 = multiply(g, g);
        poly sum(2*len);
        for (int i = 0; i < 2*len; i++) {
            int v = (i < (int)f.size() ? f[i] : 0);
            sum[i] = (v + (i < (int)g2.size() ? g2[i] : 0)) % M;
        }
        auto inv_g = poly_inv(g, 2*len);
        g = multiply(sum, inv_g);
        for (auto &x : g) x = (ll)x * inv2 % M;
        g.resize(2*len);
    }
    g.resize(m);
    return g;
}

poly poly_d(const poly& f) {
    int n = f.size();
    poly g(max<int>(0,n-1));
    for (int i = 1; i < n; i++) g[i-1] = (ll)f[i]*i % M;
    return g;
}

poly poly_int(const poly& f) {
    int n = f.size();
    poly g(n+1);

    static poly inv{0,1};
    {
        while ((int)inv.size() <= n) {
            int i = inv.size();
            inv.push_back((ll)(M - M/i) * inv[M % i] % M);
        }
    }

    for (int i = 0; i < n; i++) g[i+1] = (ll)f[i] * inv[i+1] % M;
}

```

```
    return g;
}

poly poly_ln(const poly& f, int m) {
    auto df = poly_d(f);
    auto invf = poly_inv(f, m);
    auto q = multiply(df, invf);
    q.resize(m-1);
    auto res = poly_int(q);
    res.resize(m);
    return res;
}

poly poly_exp(const poly& f, int m) {
    poly g(1,1);
    for (int len = 1; len < m; len <= 1) {
        auto g_ln = poly_ln(g, 2*len);
        poly diff(2*len);
        for (int i = 0; i < 2*len; i++) {
            diff[i] = ((i < (int)f.size() ? f[i] : 0) - g_ln[i]
                + M) % M;
        }
        diff[0] = (diff[0] + 1) % M;
        g = multiply(g, diff);
        g.resize(2*len);
    }
    g.resize(m);
    return g;
}

poly poly_pow(const poly& f, ll k, int m) {
    auto ln_f = poly_ln(f, m);
    for (int i = 0; i < m; i++) ln_f[i] = (ll)ln_f[i] * (k % (M
        -1)) % M;
    return poly_exp(ln_f, m);
}

pair<poly, poly> poly_divmod(poly a, poly b) {
    int n = a.size(), m = b.size();
    if (n < m) return {{0}, a};
    poly ra(a.rbegin(), a.rend()), rb(b.rbegin(), b.rend());
    int k = n-m+1;
    auto inv_rb = poly_inv(rb, k);
    auto rq = multiply(ra, inv_rb);
    rq.resize(k);
    poly q(rq.rbegin(), rq.rend());
    auto bq = multiply(b, q);
    poly r = a;
    for (int i = 0; i < (int)bq.size(); i++)
        r[i] = (r[i] - bq[i] + M) % M;
    r.resize(m-1);
    return {q, r};
}

int main (){
    ios::sync_with_stdio(0); cin.tie(0);
    vector<cd> a={1,2}, b={3,4};
    vector<cd> c=multiply(a,b);
    for (auto e:c) cout << e << ' ' ;
}
```

Convolutions (12)

AndConvolution.hpp

Description: Bitwise AND Convolution. Superset Zeta Transform: $A'[S] = \sum_{T \supseteq S} A[T]$. Superset Mobius Transform: $A[T] = \sum_{S \supseteq T} (-1)^{|S-T|} A'[S]$.

```
Time:  $\mathcal{O}(N \log N)$ .
"../template/Header.hpp" 7916f8, 34 lines

template<class T>
void superset_zeta(vector<T> &a){
    int n=(int)a.size();
    assert(n==(n&n));
    for(int i=1;i<n;i<=1){
        for(int j=0;j<n;j++){
            if(j&i){
                a[j^i]+=a[j];
            }
        }
    }
}

template<class T>
void superset_mobius(vector<T> &a){
    int n=(int)a.size();
    assert(n==(n&n));
    for(int i=n;i>=1;){
        for(int j=0;j<n;j++){
            if(j&i){
                a[j^i]-=a[j];
            }
        }
    }
}

template<class T>
vector<T> and_convolution(vector<T> a,vector<T> b){
    superset_zeta(a);
    superset_zeta(b);
    for(int i=0;i<(int)a.size();i++)a[i]*=b[i];
    superset_mobius(a);
    return a;
}
```

GCDConvolution.hpp

Description: GCD Convolution. Multiple Zeta Transform: $A'[n] = \sum_{n|m} A[m]$. Multiple Mobius Transform: $A[n] = \sum_{n|m} \mu(m/n) A'[m]$.

```
Time:  $\mathcal{O}(N \log \log N)$ .
"../template/Header.hpp" 7f6c2d, 34 lines

template<class T>
void multiple_zeta(vector<T> &a){
    int n=(int)a.size();
    vector<bool> is_prime(n,true);
    for(int p=2;p<n;p++){
        if(!is_prime[p])continue;
        for(int i=(n-1)/p;i>=1;i--){
            is_prime[i*p]=false;
            a[i]+=a[i*p];
        }
    }
}

template<class T>
void multiple_mobius(vector<T> &a){
    int n=(int)a.size();
    vector<bool> is_prime(n,true);
    for(int p=2;p<n;p++){
        if(!is_prime[p])continue;
        for(int i=1;i*p<n;i++){
            is_prime[i*p]=false;
            a[i]-=a[i*p];
        }
    }
}
```

```
template<class T>
vector<T> gcd_convolution(vector<T> a,vector<T> b){
    multiple_zeta(a);
    multiple_zeta(b);
    for(int i=0;i<(int)a.size();i++)a[i]*=b[i];
    multiple_mobius(a);
    return a;
}
```

LCMConvolution.hpp

Description: LCM Convolution. Divisor Zeta Transform: $A'[n] = \sum_{d|n} A[d]$. Divisor Mobius Transform: $A[n] = \sum_{d|n} \mu(n/d) A'[d]$.

```
Time:  $\mathcal{O}(N \log \log N)$ .
"../template/Header.hpp" 41fe9d, 34 lines

template<class T>
void divisor_zeta(vector<T> &a){
    int n=(int)a.size();
    vector<bool> is_prime(n,true);
    for(int p=2;p<n;p++){
        if(!is_prime[p])continue;
        for(int i=1;i*p<n;i++){
            is_prime[i*p]=false;
            a[i*p]+=a[i];
        }
    }
}

template<class T>
void divisor_mobius(vector<T> &a){
    int n=(int)a.size();
    vector<bool> is_prime(n,true);
    for(int p=2;p<n;p++){
        if(!is_prime[p])continue;
        for(int i=(n-1)/p;i>=1;i--){
            is_prime[i*p]=false;
            a[i*p]-=a[i];
        }
    }
}

template<class T>
vector<T> lcm_convolution(vector<T> a,vector<T> b){
    divisor_zeta(a);
    divisor_zeta(b);
    for(int i=0;i<(int)a.size();i++)a[i]*=b[i];
    divisor_mobius(a);
    return a;
}
```

ORConvolution.hpp

Description: Bitwise OR Convolution. Subset Zeta Transform: $A'[S] = \sum_{T \subseteq S} A[T]$. Subset Mobius Transform: $A[T] = \sum_{S \subseteq T} (-1)^{|T-S|} A'[S]$.

```
Time:  $\mathcal{O}(N \log N)$ .
"../template/Header.hpp" c58b77, 34 lines

template<class T>
void subset_zeta(vector<T> &a){
    int n=(int)a.size();
    assert(n==(n&n));
    for(int i=1;i<n;i<=1){
        for(int j=0;j<n;j++){
            if(j&i){
                a[j]+=a[j^i];
            }
        }
    }
}
```



```
template<class T>
void subset_mobius(vector<T> &a){
    int n=(int)a.size();
    assert(n==(n&-n));
    for(int i=n;i>=1;){
        for(int j=0;j<n;j++){
            if(j&i){
                a[j]-=a[j^i];
            }
        }
    }
}

template<class T>
vector<T> or_convolution(vector<T> a,vector<T> b){
    subset_zeta(a);
    subset_zeta(b);
    for(int i=0;i<(int)a.size();i++)a[i]*=b[i];
    subset_mobius(a);
    return a;
}
```

XORConvolution.hpp

Description: Bitwise XOR Convolution. Fast Walsh-Hadamard Transform:
 $A'[S] = \sum_T (-1)^{|S \& T|} A[T]$.
Time: $\mathcal{O}(N \log N)$.

../template/Header.hpp05848d, 29 lines

```
template<class T>
void fwht(vector<T> &a){
    int n=(int)a.size();
    assert(n==(n&-n));
    for(int i=1;i<n;i<=1){
        for(int j=0;j<n;j++){
            if(j&i){
                T &u=a[j^i],&v=a[j];
                tie(u,v)=make_pair(u+v,u-v);
            }
        }
    }
}

template<class T>
vector<T> xor_convolution(vector<T> a,vector<T> b){
    int n=(int)a.size();
    fwht(a);
    fwht(b);
    for(int i=0;i<n;i++)a[i]*=b[i];
    fwht(a);
    T div=T(1)/T(n);
    if(div==T(0)){
        for(auto &x:a)x/=n;
    }else{
        for(auto &x:a)x*=div;
    }
    return a;
}
```

MaxPlusConvolution.hpp

Description: Max Plus Convolution. Find $C[k] = \max_{i+j=k} \{A[i] + B[j]\}$.
Time: $\mathcal{O}(N)$.

7176a2, 94 lines

```
// SMAWCK algorithm for finding row-wise maxima.
// f(i,j,k) checks if M[i][j] <= M[i][k].
// f(i,j,k) checks if M[i][k] is at least as good as M[i][j].
// higher is better.
template<class F>
vector<int> smawck(const F &f,const vector<int> &rows,const
vector<int> &cols){
```

```
int n=(int)rows.size(),m=(int)cols.size();
if(max(n,m)<=2){
    vector<int> ans(n,-1);
    for(int i=0;i<n;i++){
        for(int j:cols){
            if(ans[i]==-1||f(rows[i],ans[i],j)){
                ans[i]=j;
            }
        }
    }
    return ans;
}
if(n<m){
    // reduce
    vector<int> st;
    for(int j:cols){
        while(true){
            if(st.empty()){
                st.emplace_back(j);
                break;
            }else if(f(rows[(int)st.size()-1],st.back(),j)){
                st.pop_back();
            }else if(st.size()<n){
                st.emplace_back(j);
                break;
            }else{
                break;
            }
        }
    }
    return smawck(f,rows,st);
}
vector<int> ans(n,-1);
vector<int> new_rows;
for(int i=1;i<n;i+=2){
    new_rows.emplace_back(rows[i]);
}
auto res=smawck(f,new_rows,cols);
for(int i=0;i<new_rows.size();i++){
    ans[2*i+1]=res[i];
}
for(int i=0,l=0,r=0;i<n;i+=2){
    if(i+l==n)r=m;
    while(r<m&&cols[r]<=ans[i+1])r++;
    ans[i]=cols[l++];
    for(;l<r;l++){
        if(f(rows[i],ans[i],cols[l])){
            ans[i]=cols[l];
        }
    }
    l--;
}
return ans;
}

template<class F>
vector<int> smawck(const F &f,int n,int m){
    vector<int> rows(n),cols(m);
    iota(rows.begin(),rows.end(),0);
    iota(cols.begin(),cols.end(),0);
    return smawck(f,rows,cols);
}
```

```
// Max Plus Convolution.
// b must be convex, i.e. b[i]-b[i-1]>=b[i+1]-b[i].
template<class T>
vector<T> max_plus_convolution_arbitrary_convex(vector<T> a,
const vector<T> &b){
```

```
if(a.empty()||b.empty())return {};
if((int)b.size()==1){
    for(auto &x:a)x+=b[0];
    return a;
}
int n=(int)a.size(),m=(int)b.size();
auto f=[&](int i,int j){
    return a[j]+b[i-j];
};
auto cmp=[&](int i,int j,int k){
    if(i<k)return false;
    if(i-j>=m)return true;
    return f(i,j)<=f(i,k);
};
auto best=smawck(cmp,n+m-1,n);
vector<T> ans(n+m-1);
for(int i=0;i<n+m-1;i++){
    ans[i]=f(i,best[i]);
}
return ans;
}
```

Various (13)

GaussianElimination.hpp

Description: Gaussian Elimination

d847fe, 45 lines

```
const double EPS = 1e-9;
const int INF = 2; // it doesn't actually have to be infinity
or a big number

int gauss (vector < vector<double> > a, vector<double> & ans) {
    int n = (int) a.size();
    int m = (int) a[0].size() - 1;

    vector<int> where (m, -1);
    for (int col=0, row=0; col<m && row<n; ++col) {
        int sel = row;
        for (int i=row; i<n; ++i)
            if (abs (a[i][col]) > abs (a[sel][col]))
                sel = i;
        if (abs (a[sel][col]) < EPS)
            continue;
        for (int i=col; i<=m; ++i)
            swap (a[sel][i], a[row][i]);
        where[col] = row;

        for (int i=0; i<n; ++i)
            if (i != row) {
                double c = a[i][col] / a[row][col];
                for (int j=col; j<=m; ++j)
                    a[i][j] -= a[row][j] * c;
            }
        ++row;
    }

    ans.assign (m, 0);
    for (int i=0; i<m; ++i)
        if (where[i] != -1)
            ans[i] = a[where[i]][m] / a[where[i]][i];
    for (int i=0; i<n; ++i) {
        double sum = 0;
        for (int j=0; j<m; ++j)
            sum += ans[j] * a[i][j];
        if (abs (sum - a[i][m]) > EPS)
            return 0;
    }
}
```



```
for (int i=0; i<m; ++i)
    if (where[i] == -1)
        return INF;
return 1;
}
```

XORBasis.hpp

Description: XOR Basis 4f2ef8, 33 lines

```
template<int BIT>
struct XOR_basis {
    vector<ll> basis;
    int mSize = 0;
    XOR_basis() { basis.assign(BIT, 0); }
    //Insert v into basis, if v is already in the span of the
    //basis, do nothing. O(BIT)
    void insert(ll v) {
        for (int i = BIT-1; i >= 0; --i) {
            if (!(v>>i & 1)) continue;
            if (basis[i]) v ^= basis[i];
            else {basis[i] = v, mSize++; return; }
        }
    }
    int size() { return mSize; }
    //Perform row reduction to make basis in reduced row echelon
    //form in O(BIT^2)
    void reduce() {
        for (int i = BIT-1; i >= 0; --i) {
            for (int j = i-1; j >= 0; --j) {
                if (!basis[j]) continue;
                if (basis[i]>>j & 1) basis[i] ^= basis[j];
            }
        }
    }
    //Check whether v is in span of the current basis in O(BIT)
    bool in_span(ll v) {
        for (int i = BIT-1; i >= 0; --i) {
            if (!(v>>i & 1)) continue;
            v ^= basis[i];
        }
        return (v == 0);
    }
    ll &operator[] (int i) { return basis[i]; }
};
```

RangeXor.hpp

Description: find all range of x such that l <= x xor p < r. cc7fb9, 18 lines

```
template<class F>
void range_xor(ll p,ll l,ll r,const F &query){
    for(int i=0;i<60;i++){
        if(l==r)break;
        ll b=1LL<<i;
        if(l&b){
            query(l^p,(l^p)+b);
            l+=b;
        }
        if(r&b){
            r-=b;
            query(r^p,(r^p)+b);
        }
        if(p&b){
            p^=b;
        }
    }
}
```

13.1 Optimization tricks

__builtin_ia32_ldmxcsr(40896); disables denormals (which make floats 20x slower near their minimum value).

13.1.1 Bit hacks

- $x \& -x$ is the least bit in x .
- `for (int x = m; x; ) { --x &= m; ... }` loops over all subset masks of m (except m itself).
- $c = x \& -x, r = x + c; ((r^x) >> 2) / c$ | r is the next number after x with the same number of bits set.
- `rep(b,0,K) rep(i,0,(1 << K))`
if $(i \& 1 << b)$ $D[i] += D[i^ (1 << b)]$;
computes all sums of subsets.

13.1.2 Pragmas

- `#pragma GCC optimize ("Ofast")` will make GCC auto-vectorize loops and optimizes floating points better.
- `#pragma GCC target ("avx2")` can double performance of vectorized code, but causes crashes on old machines.
- `#pragma GCC optimize ("trapv")` kills the program on integer overflows (but is really slow).